

Malthus Meets Romer: Population Growth Under Information Frictions

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Abstract

Households disagree about whether larger populations enhance productivity (Romerians) or diminish it (Malthusians). I embed this disagreement into a Barro–Becker fertility model with global games. Wrong beliefs generate distinct pathologies depending on which view is correct. In Romerian worlds, Malthusians cause prolonged stagnation through an *informational poverty trap*. In Malthusian worlds, Romerians generate non-monotone dynamics, including boom-bust cycles. Optimal communication policy by the social planner is truthful under a dynastic welfare criterion but shaded under utilitarian ones. Survey data confirm that Malthusian beliefs significantly correlate with lower fertility.

JEL Classification: D83, J13, O40, C72

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1 Introduction

How many children should a family have? The answer turns on what one believes about the economic consequences of population size. A parent who thinks, with [Romer \(1990\)](#), that more people generate more ideas and faster growth may view an additional child as contributing to future prosperity. A parent who fears, with [Malthus \(1798\)](#), that more people strain finite resources may view the same child as hastening collective impoverishment. Under intergenerational altruism, these beliefs translate directly into fertility choices, and the resulting population dynamics feed back into productivity outcomes that either validate or refute them. This paper studies the consequences of that feedback loop when households hold heterogeneous beliefs about the population–productivity relationship.

I formalize the idea in an overlapping generations model in which households make fertility decisions under uncertainty about a single structural parameter: the scale effect of population on productivity growth. Households observe the same aggregate data but interpret it through different priors. “Romerians” believe that larger populations accelerate innovation and growth; “Malthusians” believe that larger populations depress living standards. Beliefs are transmitted across generations through a combination of cultural inheritance and Bayesian updating from observed outcomes. Following [Becker \(1960\)](#), children are costly in time rather than in goods: a parent devotes a fraction $\psi(n)$ of a unit time endowment to child-rearing and supplies the remainder to the labor market. Productivity itself evolves according to a multiplicative log-normal process whose drift depends on population, so today’s fertility shapes tomorrow’s productivity through the size of the next generation. Together, these ingredients yield a tractable framework in which beliefs, fertility, and growth co-evolve, and in which the analytical results below follow from a log-linear approximation around a balanced-growth path.

The model rests on a single behavioral premise: beliefs about population’s economic consequences shape fertility. Before drawing out theoretical implications, I establish this premise empirically. Using data from the General Social Survey (GSS), I document robust negative correlations between Malthusian beliefs—measured by agreement with the statement that Earth cannot continue to support population growth—and both ideal and actual fertility. Individuals with Malthusian beliefs report ideal fertility about 0.27 children lower than non-Malthusians and have 0.29 fewer children, gaps of roughly 11 and 16 percent of the respective sample means. The relationships persist after controlling for age, education, sex, and survey year, and they show up across demographic groups. Notably, the effect on *actual* fertility is at least as large as the effect on stated preferences, which suggests that beliefs translate into consequential behavior rather than cheap talk.

With this empirical foundation in place, I turn to the theoretical analysis. The strategic structure of fertility partitions the state space into three regions with distinct equilibrium properties. In the *Malthusian-dominated region*, where the share of Romerians is below a threshold ϕ^* determined by the calibration, the aggregate best response is downward-sloping and the equilibrium is unconditionally unique. In the *contraction region* between ϕ^* and a second threshold ϕ^{**} , the aggregate best response satisfies $0 \leq \rho(\phi) < 1$, and uniqueness follows from a contraction argument. In the *multiplicity region* above ϕ^{**} , strategic complementarity among Romerians dominates, multiple equilibria coexist, and selection becomes a substantive question rather than a refinement of a refinement. Whether the multiplicity region is empirically active depends on the strength of scale effects.¹ The paper handles all three regions and uses a global-games refinement à la Frankel et al. (2003) to select a risk-dominant equilibrium in the multiplicity region.

Within this equilibrium structure, the main theoretical contribution is to show that wrong beliefs generate qualitatively different pathologies depending on which worldview is correct. The asymmetry arises from a feature of Bayesian learning about scale effects: the informativeness of productivity signals depends on population size. When population sits near the reference level $\bar{N}$, productivity growth reveals almost nothing about whether scale effects are positive or negative. Learning therefore requires population to deviate substantially from this uninformative region, but such deviations require precisely the optimistic beliefs that pessimism suppresses. This interaction between beliefs, fertility, and the informativeness of data is the engine driving the paper’s main results.

Consider first a Romerian world, where scale effects are truly positive. Here, Malthusian minorities do not merely slow convergence to efficiency—they can trap the economy in prolonged stagnation. Pessimistic beliefs suppress fertility, keeping population near $\bar{N}$. A population near $\bar{N}$ generates uninformative signals about the population–productivity relationship, and uninformative signals fail to correct pessimistic beliefs. The trap is self-reinforcing: the economy remains stuck not because Malthusianism is correct, but because population never deviates enough from $\bar{N}$ to reveal that it is wrong. I call this the *informational poverty trap*. While correct beliefs eventually prevail almost surely, the expected escape time can exceed any relevant historical horizon.

Now consider the mirror case: a Malthusian world, where scale effects are truly negative. Here the dynamics differ starkly. Demographic selection always favors optimists, who have more children regardless of whether their optimism is warranted. Bayesian learning, by

¹Under Jones-style semi-endogenous growth calibrations (Jones, 1995; Bloom et al., 2020), the multiplicity region is empty; under Kremer-style or Galor–Weil-style calibrations (Kremer, 1993; Galor, 2011), it is occupied at high ϕ .

contrast, favors pessimists when population growth leads to poor outcomes. The two forces pull the Romerian share ϕ_t in opposite directions, and the balance between them itself depends on the state. The paper’s second main theoretical result formalizes this opposition: in any Malthusian world, the expected drift in ϕ_t cannot be uniformly signed, because the learning channel vanishes at $N_t = \bar{N}$ (where selection dominates) and grows quadratically in $\ln(N_t/\bar{N})$ thereafter. Monotone convergence is therefore ruled out, and the system inherits the potential for non-monotone dynamics, including the boom-bust cycles familiar from pre-industrial demography. During expansions, optimists outbreed pessimists and population grows. Eventually the economy hits constraints, outcomes deteriorate, learning corrects beliefs toward Malthusianism, and fertility falls. Recovery begins, and the pattern can repeat.

Taken together, the two cases reveal a sharp duality: *in Romerian worlds, Malthusians cause stagnation; in Malthusian worlds, Romerians generate non-monotone dynamics, including cycles.* Wrong beliefs are always costly, but the nature of the cost depends on the direction of the error.

This duality is more than a theoretical curiosity. It offers a unified lens for interpreting demographic history—though one that admits two competing readings of the pre-industrial Malthusian era. The conventional reading is that the pre-industrial world was genuinely Malthusian: scale effects were negative, and the cycles documented by economic historians (Clark, 2007; Voigtländer and Voth, 2013) reflect the boom-bust dynamics of a true Malthusian regime. On this view, the Industrial Revolution marked a genuine technological regime change. The alternative reading, made possible by the informational poverty trap, is that the world was always Romerian, but humanity was stuck in a trap where population never deviated enough from the reference to generate the information needed to escape pessimism. On this view, the Industrial Revolution was not a technological regime change but an *informational escape*: some combination of circumstances pushed population into the informative region, beliefs updated toward Romerianism, and the trap dissolved. Both readings are consistent with the historical evidence, and distinguishing them carries profound implications for understanding the modern fertility decline—which the model can interpret as a partial drift back toward the trap, an efficient response to a renewed Malthusian regime, or a quantity-quality substitution unrelated to scale effects.

The theoretical results naturally raise a policy question: if wrong beliefs can trap economies in stagnation or generate damaging cycles, can public communication help? Answering this question requires first confronting what “help” even means when the population itself is endogenous to the policy. As Golosov et al. (2007) emphasize, standard welfare criteria become

problematic when the set of agents depends on policy.² I analyze optimal policy under two welfare criteria that span a substantively distinct normative spectrum: a dynastic criterion that respects Barro–Becker preferences, and a total utilitarian criterion that aggregates utility across all individuals who exist under the true law of motion. The contrast suffices to bring out the genuinely distinct policy implications. A dynastic planner is fundamentally an information provider; a total utilitarian planner is an externality corrector. The former is indifferent to belief correction beyond information provision; the latter has a stake in both truth and population. An average-utilitarian benchmark sits between them and is treated as a corollary.

Optimal announcement policy differs substantially across welfare criteria. A dynastic planner favors truthful communication: under subjective preferences, any informative signal is welfare-improving and there is no externality to correct. A total utilitarian planner shades announcements toward optimism (in a Romerian world) by an amount that depends on the size of the externality, the deviation of population from the reference level, and the dispersion of priors. An average-utilitarian planner shades less than the total utilitarian but more than the dynastic, because it internalizes the per-capita scale-effect spillover but not the population-size motive. Normative conclusions about the costs of wrong beliefs thus depend critically on the welfare criterion employed, and the contrast across criteria is itself a substantive result of the policy analysis.

How announcements operate depends on where the economy sits in the state space. In the contraction region, where the equilibrium is unique, announcements act purely as an information-provision instrument: they shift households’ subjective posteriors toward truth (or, by design, away from it when an externality is being corrected). In the multiplicity region, announcements additionally play a selection-shifting role: by coordinating beliefs about which equilibrium will be played, they can move the economy from a low-fertility selection to a high-fertility selection (or vice versa). The trap region sits inside the contraction region, so the trap intuition for policy is unaffected by the multiplicity question. The multiplicity region is where the global-games refinement earns its keep, and where the planner’s announcement combines information provision with selection shifting.

Beyond the choice of welfare criterion, the analysis of public information itself reveals a novel asymmetry. In standard coordination games, public information can be harmful by inducing excessive coordination on potentially incorrect signals (Morris and Shin, 2002). In my setting, the welfare implications depend on where the economy sits. When the economy is trapped in the uninformative region, public information is most valuable: a credible

²The axiomatic foundations of population ethics, including the role of critical-level utilitarianism in addressing the “Repugnant Conclusion” of Parfit (1984), are developed in Blackorby et al. (2005).

announcement can coordinate beliefs and push population into the informative region, initiating escape. But this is precisely when the planner faces maximum uncertainty—the same uninformative signals that trap households also limit what the planner can learn. The planner’s dilemma is sharpest when intervention matters most.

This tension yields a policy prescription that differs from standard results. In the informational poverty trap, even a moderately noisy announcement can improve welfare by breaking self-reinforcing pessimism. In a Malthusian world with non-monotone dynamics, by contrast, announcements require higher precision to avoid amplifying volatility. The heterogeneous strategic interaction—Romerians exhibiting strategic complementarity and Malthusians strategic substitutability—provides a natural stabilizer that makes public information less dangerous than in pure-complementarity environments, though the risk does not vanish.

Related Literature

This paper connects several strands of literature: endogenous fertility and population, heterogeneous beliefs in macroeconomics, global games, and cultural transmission.

The foundational work on endogenous fertility includes [Becker \(1960\)](#), who introduced both the quantity-quality tradeoff and the time-cost-of-children formulation that I adopt, and [Barro and Becker \(1989\)](#), who embedded fertility choice in a dynastic optimization framework. The quantitative-fertility literature has refined the calibration of preferences and time costs; [de la Croix and Doepke \(2003\)](#), in particular, establish that the time cost of one child is approximately fifteen percent of a parent’s time endowment, which I use to calibrate the curvature parameter of the time-cost function. Subsequent contributions enrich this framework with human capital accumulation ([Galor and Weil, 2000](#)), mortality transitions ([Kalemli-Ozcan, 2003](#)), and intergenerational transfers ([Lee, 2003](#)). The unified growth theory of [Galor \(2011\)](#) synthesizes these elements to explain the demographic transition from Malthusian stagnation to modern growth. My contribution is distinct: I introduce belief heterogeneity as a driver of fertility differentials and population dynamics, rather than focusing on the evolution of technology or human capital.

The debate over population scale effects has a long history. [Malthus \(1798\)](#) articulated the view that population growth outstrips resource growth, leading to immiseration. [Romer \(1990\)](#) and [Kremer \(1993\)](#) formalized the opposing view that larger populations generate more ideas, accelerating technological progress. [Jones \(1995\)](#) and [Jones \(2022\)](#) provide frameworks that nest both perspectives, with semi-endogenous growth implying modest positive scale effects, and [Bloom et al. \(2020\)](#) document declining research productivity that sharpens empirical estimates of effective scale-effect parameters. Empirical work by [Ashraf and Galor](#)

(2013) finds evidence for positive scale effects over very long horizons, while [Voigtländer and Voth \(2013\)](#) documents Malthusian dynamics in pre-industrial Europe. I do not take a stand on which view is correct; instead, I study the consequences of households holding different views.

The literature on heterogeneous beliefs has grown substantially. In macroeconomics, [Angeletos and La'O \(2013\)](#) and [Angeletos et al. \(2018\)](#) examine how dispersed information affects business cycles and sentiments; [Bordalo et al. \(2018\)](#) develop diagnostic expectations with belief overreaction; and [Malmendier and Nagel \(2011\)](#) document that personal experiences shape macroeconomic expectations. In finance, [Simsek \(2013\)](#), [Geanakoplos \(2010\)](#), and [Scheinkman and Xiong \(2003\)](#) study asset pricing under disagreement. My paper differs by modeling disagreement about structural parameters governing long-run relationships rather than beliefs about current states or short-run forecasts.

The global games literature, pioneered by [Carlsson and Van Damme \(1993\)](#) and developed by [Morris and Shin \(1998, 2003\)](#) and [Frankel et al. \(2003\)](#), provides tools for analyzing games with strategic complementarities under incomplete information. I adapt these techniques to a setting that mixes strategic complementarities (among Romerians) and strategic substitutabilities (among Malthusians). This combination creates analytical challenges but also generates richer dynamics than standard global games applications such as currency crises ([Morris and Shin, 1998](#)) or bank runs ([Goldstein and Pauzner, 2005](#)). The selection theorem of [Frankel et al. \(2003\)](#) for general supermodular games extends, after a reduction to the Romerian subgame, to the Romerian-dominated multiplicity region of my model, and I use it to select a risk-dominant equilibrium there.

The literature on cultural transmission, developed by [Cavalli-Sforza and Feldman \(1981\)](#), [Bisin and Verdier \(2001, 2011\)](#), and applied by [Doepke and Zilibotti \(2019\)](#) to parenting styles, provides frameworks for understanding how preferences and beliefs persist across generations. I incorporate cultural transmission as one channel through which beliefs evolve, alongside Bayesian learning from observed outcomes.

The analysis of public information in coordination games relates to [Morris and Shin \(2002\)](#), who show that in games with strategic complementarities, public information can be harmful if it crowds out private information. In my setting, the welfare implications are more nuanced because of the heterogeneous strategic interaction and the state-dependent value of coordination: public information is most valuable in the trap, where it is also most uncertain.

An important related literature addresses welfare evaluation with endogenous population. [Goloso et al. \(2007\)](#) provide a systematic analysis of efficiency concepts when fertility is a choice variable, showing that standard welfare theorems fail under total or average utilitari-

anism but that a modified Pareto criterion can restore efficiency of competitive equilibria. I build on these insights by analyzing how dynastic and total utilitarian welfare criteria affect optimal communication policy in a setting with belief heterogeneity.

Finally, the paper relates to work on poverty traps and multiple equilibria in development economics, such as [Azariadis and Drazen \(1990\)](#) on traps from threshold externalities and [Banerjee and Newman \(1993\)](#) on occupational choice with wealth constraints. The informational poverty trap I identify differs from these mechanisms: it arises not from technological nonconvexities or credit constraints but from the endogenous uninformativeness of signals when the economy remains in a particular region of the state space.

The paper proceeds as follows. Section 2 presents the model. Section 3 provides empirical evidence validating the central prediction that Malthusian beliefs reduce fertility. Section 4 analyzes strategic complementarity and substitutability, partitions the state space into three regions, and applies the global-games selection argument to the multiplicity region. Section 5 examines long-run dynamics, including the informational poverty trap and the non-monotone dynamics of Malthusian worlds. Section 6 discusses the alignment between theoretical predictions and demographic history. Section 7 analyzes the value of public information and characterizes optimal communication policy. Section 8 concludes. All proofs and auxiliary results are collected in the Appendix.

2 Model

I develop a discrete-time overlapping generations model with dynastic altruism. The economy is populated by a continuum of households who make fertility decisions and differ in their beliefs about the structural relationship between population and productivity growth.

2.1 Environment

Time is discrete and indexed by $t = 0, 1, 2, \dots$. Each period, a generation of adults makes economic decisions; their children become adults in the following period.

Time Endowment and Cost of Children. Each adult is endowed with one unit of time. An adult who chooses fertility n_{it} spends a fraction $\psi(n_{it}) \in [0, 1)$ of that time on child-rearing and supplies the remainder $1 - \psi(n_{it})$ to the labor market. Child-rearing therefore competes directly with market work, and the implicit price of an additional child is the forgone labor income. This time-cost specification, originally articulated by [Becker \(1960\)](#), has the important property that the resource constraint on consumption is multiplicative rather than additive: a household's consumption is its labor income net of the time spent on

children, not its labor income minus a goods expenditure. The multiplicative structure plays a central analytical role in what follows by making the household value function tractable to a first-order log-linear approximation.

Production. A single final good is produced using effective labor according to the linear technology

$$Y_t = A_t L_t, \quad L_t = N_t \cdot \int_0^1 (1 - \psi(n_{it})) di, \quad (1)$$

where Y_t is output, A_t is total factor productivity, N_t is the adult population, and L_t is aggregate effective labor supply. I interpret A_t broadly as an “effective” TFP that incorporates any potential dilution effects from fixed factors. The competitive wage equals the marginal product of effective labor, $w_t = A_t$, so a type- j household with fertility $n_{j,t}$ supplies $1 - \psi(n_{j,t})$ units of labor and earns $A_t(1 - \psi(n_{j,t}))$. With no other income or expenditure, consumption is

$$c_{j,t} = A_t(1 - \psi(n_{j,t})), \quad (2)$$

and aggregate per-capita output and consumption coincide automatically.

TFP Dynamics. TFP evolves according to a multiplicative log-normal process,

$$A_{t+1} = A_t \cdot \exp(\bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}), \quad \eta_{t+1} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2), \quad (3)$$

where $\bar{g} \geq 0$ is baseline log-productivity growth, $\theta^* \in \mathbb{R}$ is the *scale-effect parameter*, $\bar{N} > 0$ is an empirically anchored *reference population*, and η_{t+1} is an i.i.d. structural productivity shock independent of all histories. Taking logs gives the exact identity

$$\ln A_{t+1} = \ln A_t + \bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}, \quad (4)$$

which is linear in $\ln A_t$, $\ln(N_t/\bar{N})$, and η_{t+1} . The technology side of the model contributes *no* Taylor remainder: any approximation in what follows enters only through the optimal fertility policy and the value-function conjecture below.³ Throughout, I write $g(N_t; \theta) \equiv \bar{g} + \theta \ln(N_t/\bar{N})$ for the productivity-growth function.

When $\theta^* > 0$, larger populations generate faster productivity growth (Romer); when $\theta^* < 0$, larger populations slow productivity growth (Malthusian dynamic congestion). When $N_t = \bar{N}$, the deterministic component of productivity growth equals $\bar{g}$ regardless of θ^* . Observations near $N_t = \bar{N}$ are therefore *uninformative* about the scale-effect parameter,

³The reference population $\bar{N}$ appears explicitly, so the model is invariant under rescaling of the population unit: only $\ln(N_t/\bar{N})$ matters for the dynamics.

even though they remain noisy through η_{t+1} . This feature is central to the informational poverty trap.

The parameter $\theta < 0$ here represents *dynamic* congestion or environmental drag—the marginal effect of population on the growth rate of productivity—rather than classical Malthusian fixed-factor dilution operating on the level of output.⁴

Households and Beliefs. The economy contains a unit mass of dynastic households indexed by $i \in [0, 1]$. Each household holds a belief θ_i about the true value of the scale-effect parameter.

Assumption 1 (Belief Types). *A fraction $\phi_t \in [0, 1]$ of households are Romerians with belief $\theta_R > 0$; the remaining $1 - \phi_t$ are Malthusians with belief $\theta_M < 0$.*

A “Romerian world” has $\theta^* > 0$; a “Malthusian world” has $\theta^* < 0$. Households do not know θ^* with certainty but hold a common prior $\pi_t \in [0, 1]$ that the world is Romerian and update via Bayes’ rule. The distribution of beliefs ϕ_t is common knowledge in each period, but individual household types are private information.

Time-Cost Function. I impose the following regularity conditions.

Assumption 2 (Time Cost of Children). *The time-cost function $\psi : \mathbb{R}_+ \rightarrow [0, 1]$ satisfies (i) $\psi(0) = 0$; (ii) $\psi'(n) > 0$ for $n > 0$ and $\psi''(n) \geq 0$ for all $n \geq 0$; (iii) $\psi(n) < 1$ for finite n with $\lim_{n \rightarrow \infty} \psi(n) = 1$; and (iv) the marginal time cost $\psi'(n)/(1 - \psi(n))$ is strictly increasing in n with $\lim_{n \rightarrow \infty} \psi'(n)/(1 - \psi(n)) = \infty$.*

For analytical tractability, I use the exponential-quadratic specification

$$\psi(n) = 1 - e^{-\frac{\gamma}{2}n^2}, \quad \gamma > 0, \quad (5)$$

which yields the convenient identity $\psi'(n)/(1 - \psi(n)) = \gamma n$. This identity exactly mirrors the linear marginal-cost structure of a goods-cost specification while keeping consumption bounded between zero and A_t . Following the calibration of [de la Croix and Doepke \(2003\)](#), the time cost of one child is approximately fifteen percent of a parent’s time endowment, which under (5) pins down $\gamma = -2 \ln(0.85) \approx 0.325$. I take this value as the baseline calibration.

⁴Capturing the latter would require an additional production margin (a fixed factor in the production function), which is straightforward but distinct from the channel modeled here. The current setup is best understood as a reduced-form representation of dynamic effects of population on innovation, in the spirit of [Romer \(1990\)](#) and [Jones \(1995\)](#) on the positive- θ side, and of environmental-drag and congestion models on the negative- θ side.

Preferences. Households have dynastic preferences:

$$U_{it} = u(c_{it}) + \beta n_{it} \mathbb{E}_{it} [V_{i,t+1}], \quad (6)$$

where $u(c) = \ln c$, $\beta \in (0, 1)$ is the dynastic discount factor, and $V_{i,t+1}$ is the continuation value for the dynasty starting next period. Throughout, I assume equilibrium fertility satisfies $\beta n_j^* < 1$ so that the dynastic value function is well defined. I set $\beta = 0.30$, interpreted as a structural per-generation altruism parameter rather than a compounded annual discount factor.⁵ Substituting $c_{it} = A_t(1 - \psi(n_{it}))$ gives the additive decomposition

$$\ln c_{it} = \ln A_t + \ln(1 - \psi(n_{it})). \quad (7)$$

The first term depends only on aggregate productivity, the second only on own fertility through the time cost. This separation is the analytical advantage of the time-cost specification: $\ln A_t$ carries dynamic, aggregate information, while $\ln(1 - \psi(n_{it}))$ carries the static individual choice.

Population Dynamics. Aggregate population evolves according to

$$N_{t+1} = \bar{n}_t \cdot N_t, \quad \bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}, \quad (8)$$

where $n_{R,t}$ and $n_{M,t}$ denote the fertility choices of Romerians and Malthusians, respectively.

2.2 Belief Formation and Intergenerational Transmission

Beliefs evolve through cultural transmission and Bayesian learning.

Cultural Transmission. Children initially adopt their parents' beliefs with probability $\lambda \in [0, 1]$ and learn from observed outcomes with probability $1 - \lambda$.

Assumption 3 (Cultural Transmission). *A child born to a type- j household adopts type j with probability λ and forms beliefs through learning with probability $1 - \lambda$.*

⁵This value sits at the lower end of the range used in the quantitative Barro–Becker literature, which calibrates the dynastic discount factor directly to fertility, savings, and intergenerational-transfer moments. [Manuelli and Seshadri \(2009\)](#) estimate altruism parameters in this range when matching cross-country fertility differences; [Lee and Seshadri \(2019\)](#) use a comparable calibration in their analysis of intergenerational economic-status transmission; and [Daruich \(2018\)](#) adopts a similar value in an endogenous-fertility framework with early childhood investment. The value $\beta = 0.30$ is conservative for the present analysis: a higher altruism parameter would amplify the strategic complementarity slope d_R in (18) and *enlarge* the multiplicity region of Theorem 1, making the selection results more rather than less relevant.

When $\lambda = 1$ beliefs are perfectly persistent across generations; when $\lambda = 0$ beliefs are formed entirely through learning.

Bayesian Learning. Children who do not inherit parental beliefs form beliefs by Bayes' rule. Under the multiplicative log-normal TFP law of motion (3), observed log-productivity growth is

$$s_t \equiv \ln(A_{t+1}/A_t) = \bar{g} + \theta^* \ln(N_t/\bar{N}) + \eta_{t+1}, \quad \eta_{t+1} \sim \mathcal{N}(0, \sigma_\eta^2). \quad (9)$$

Given s_t and N_t , the likelihood under belief θ is

$$\ell(s_t \mid \theta, N_t) = \frac{1}{\sqrt{2\pi}\sigma_\eta} \exp\left(-\frac{(s_t - \bar{g} - \theta \ln(N_t/\bar{N}))^2}{2\sigma_\eta^2}\right), \quad (10)$$

and Bayes' rule gives the posterior

$$\pi_{t+1} = \frac{\pi_t \cdot \ell(s_t \mid \theta_R, N_t)}{\pi_t \cdot \ell(s_t \mid \theta_R, N_t) + (1 - \pi_t) \cdot \ell(s_t \mid \theta_M, N_t)}. \quad (11)$$

Children who learn adopt Romerian beliefs with probability π_{t+1} and Malthusian beliefs with probability $1 - \pi_{t+1}$.

Signal Informativeness. The Kullback–Leibler divergence between the signal distributions under the two beliefs is

$$\mathcal{I}_t \equiv \text{KL}(\ell(\cdot \mid \theta_R, N_t) \parallel \ell(\cdot \mid \theta_M, N_t)) = \frac{(\theta_R - \theta_M)^2 [\ln(N_t/\bar{N})]^2}{2\sigma_\eta^2}. \quad (12)$$

When $N_t = \bar{N}$, $\mathcal{I}_t = 0$: the signal is completely uninformative about θ^* . As $|\ln(N_t/\bar{N})|$ grows, the signal becomes increasingly informative. This dependence of informativeness on the deviation of population from its reference is the source of the informational poverty trap and, in Malthusian worlds, of the opposing forces analyzed in Section 5.

Evolution of Belief Distribution. Combining cultural transmission and learning, the Romerian share evolves as

$$\phi_{t+1} = \lambda \phi_t^{\text{birth}} + (1 - \lambda) \pi_{t+1}, \quad (13)$$

where $\phi_t^{\text{birth}} = \phi_t n_{R,t} / \bar{n}_t$ is the fraction of children born to Romerians.

2.3 Equilibrium

Definition 1 (Equilibrium). *An equilibrium consists of fertility policies $n_R(A_t, N_t, \phi_t, \pi_t)$, $n_M(A_t, N_t, \phi_t, \pi_t)$; aggregate fertility $\bar{n}(A_t, N_t, \phi_t, \pi_t)$; and laws of motion for $(A_{t+1}, N_{t+1}, \phi_{t+1}, \pi_{t+1})$ given by (3), (8), (13), and (11), such that (a) each household's fertility choice maximizes (6) subject to (2) given subjective beliefs and the aggregate fertility function; (b) expectations about aggregate fertility are consistent with the equilibrium aggregate; and (c) the actual laws of motion use the true θ^* and realized η_{t+1} .*

The Bellman equation below treats a type- j parent's continuation value as the value function of a type- j child, even though the child may switch with probability $1 - \lambda$.⁶

2.4 Household Optimization

The Bellman for a type- j household reads

$$V_j(A, N, \phi, \pi) = \max_{n \geq 0} \{ \ln A + \ln(1 - \psi(n)) + \beta n \cdot \mathbb{E}_j[V_j(A', N', \phi', \pi')] \}. \quad (14)$$

I work with a first-order log-linear approximation of V_j around a balanced-growth reference point. Specifically, I conjecture

$$V_j(A, N, \phi, \pi) \approx v_j(\phi, \pi) + \xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N}), \quad (15)$$

where the coefficients $(\xi_{A,j}, \xi_{N,j})$ and the residual $v_j(\phi, \pi)$ are determined by coefficient matching at the reference point. The technology side of the Bellman is exactly linear in $(\ln A, \ln(N/\bar{N}))$ under the multiplicative log-normal specification, so any deviation from (15) comes only from the dependence of optimal fertility n_j^* on the state through the FOC (16) below. Treating n_j^* as locally constant in $(\ln A, \ln(N/\bar{N}))$ at the reference point yields the closed-form coefficients reported in Lemma 1.

The first-order condition for fertility is

$$\frac{\psi'(n_j)}{1 - \psi(n_j)} = \beta \mathbb{E}_j[V_j(A', N', \phi', \pi')], \quad (16)$$

which under (5) simplifies to $\gamma n_j = \beta \mathbb{E}_j[V_j]$.

Lemma 1 (Equilibrium Fertility, Local Coefficients). *Under Assumptions 1–3 and the local log-linear conjecture (15), the value function coefficients evaluated at the balanced-growth*

⁶An exact formulation would replace $\mathbb{E}_j[V_j]$ by $\lambda \mathbb{E}_j[V_j] + (1 - \lambda) \mathbb{E}_j[V_{learn}]$, where V_{learn} is a type-mixture continuation expected by the parent. The two formulations coincide when λ is close to 1 or when V_R and V_M are close in the relevant region; both conditions hold near the steady states studied below.

reference are

$$\xi_{A,j} = \frac{1}{1 - \beta n_j^*}, \quad \xi_{N,j} = \frac{\beta n_j^* \theta_j}{(1 - \beta n_j^*)^2}. \quad (17)$$

Equilibrium fertility satisfies a fixed-point equation in n_j^* and can be written as $n_j^* = n^* h(\theta_j; \bar{n})$ with h increasing in θ . Moreover, $n_R^* > n_M^*$ for all equilibrium values of $\bar{n}$.

The proof, including a careful statement of the local nature of (17), is in Appendix A. The economic insight is that households who believe population enhances productivity (Romerians) perceive a higher return to children and choose higher fertility. This testable prediction motivates the empirical analysis in the next section.

3 Empirical Evidence: Beliefs and Fertility

A central prediction of the model is that Malthusian beliefs reduce fertility. Lemma 1 establishes that households who believe population diminishes productivity choose lower fertility than those who believe it enhances productivity. This section examines empirical support using survey data on beliefs and fertility.

3.1 Data

I use data from the General Social Survey (GSS), a nationally representative survey of U.S. adults conducted by NORC at the University of Chicago. The key belief variable comes from the question: “The earth cannot continue to support population growth at its present rate.” Respondents indicate agreement on a five-point scale, and I classify those who agree or strongly agree as holding *Malthusian beliefs*. This question was asked in the 2000 and 2010 waves.

I examine two fertility measures: stated preferences (“ideal number of children”) and realized fertility (actual number of children). Examining both lets me test whether beliefs shape intentions and outcomes. The sample contains 1,822 observations with valid responses to the belief and fertility questions and to demographic controls.

3.2 Descriptive Patterns

Table 1 presents summary statistics. Mean ideal fertility is 2.50 (s.d. 0.87) and mean actual fertility is 1.81 (s.d. 1.64). Approximately 55 percent of respondents hold Malthusian beliefs.

Figure 1 displays mean ideal and actual fertility by Malthusian belief intensity. There is a clear monotonic negative relationship between belief strength and both fertility measures. Respondents who strongly disagree report an average ideal of 2.98 children and have 1.76

Table 1. Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
Ideal number of children	1,822	2.50	0.87	0	7
Actual number of children	1,822	1.81	1.64	0	8
Malthusian belief (binary)	1,822	0.55	0.50	0	1
Malthusian belief (1–5 scale)	1,822	3.44	1.03	1	5
Age	1,822	46.8	17.3	18	89
Years of education	1,822	13.3	2.96	0	20
Female	1,822	0.56	0.50	0	1

Notes: Data from General Social Survey, 2000 and 2010 waves. Malthusian belief (binary) equals one if respondent agrees or strongly agrees that Earth cannot continue to support population growth.

children, while those who strongly agree report an average ideal of 2.24 children and have 1.52 children.

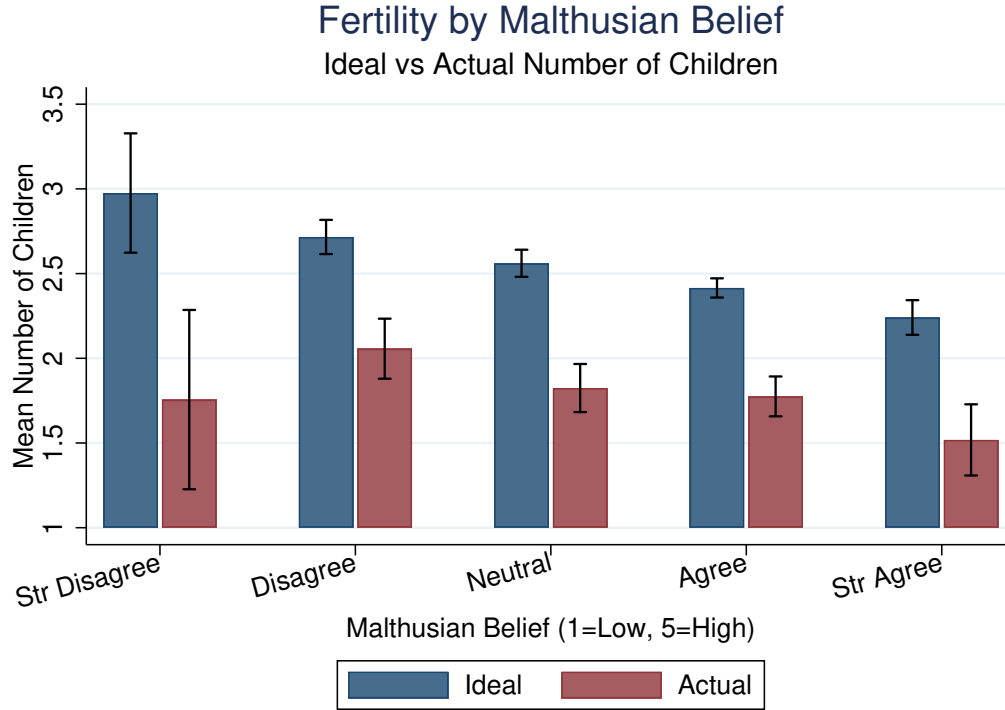


Figure 1. Fertility by Malthusian Belief Intensity

Notes: Data from General Social Survey, 2000 and 2010 waves (N = 1,822). Bars show mean ideal (navy) and actual (maroon) number of children for each level of the Malthusian belief scale. Error bars indicate 95% confidence intervals.

A natural concern is that this relationship is driven by education. Figure 2 addresses this

concern by displaying mean ideal fertility by education group and belief type. Within every education category, Malthusians report lower ideal fertility than non-Malthusians. The gap ranges from 0.20 children among college graduates to 0.45 children among those with less than high school education.

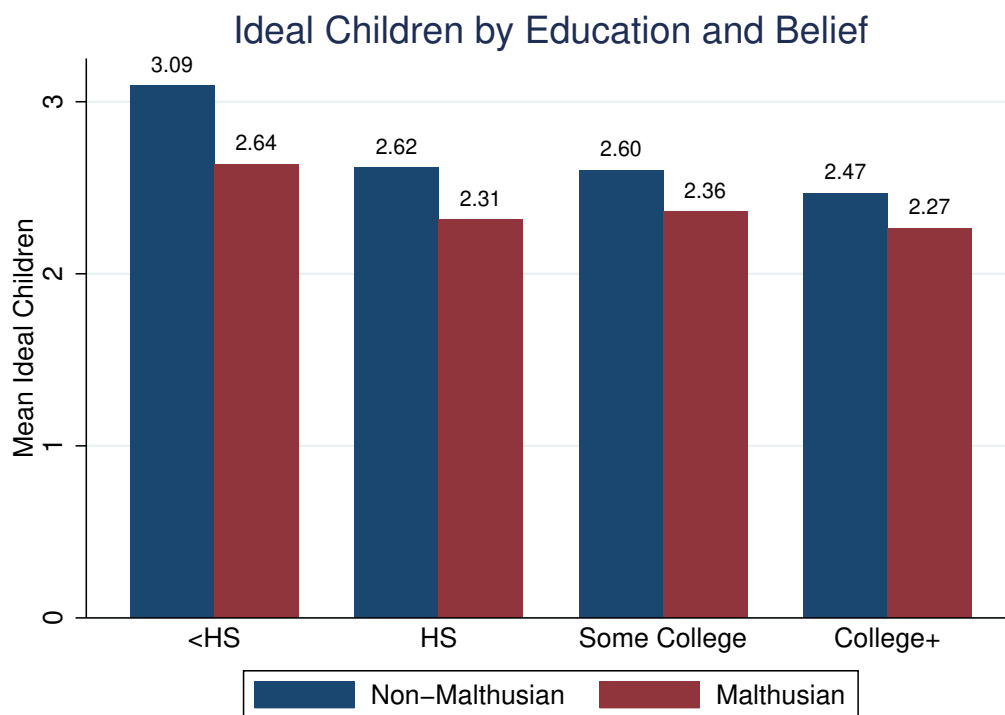


Figure 2. Ideal Number of Children by Education and Malthusian Belief

Notes: Bars show mean ideal number of children for non-Malthusians (navy) and Malthusians (maroon) within each education category.

3.3 Regression Analysis

Table 2 presents linear regressions of both ideal and actual fertility on Malthusian beliefs with demographic controls.⁷

Malthusians report ideal fertility 0.27 children lower than non-Malthusians (column 1), about 11 percent of the sample mean. The coefficient on actual fertility is -0.29 (column 3), implying Malthusians have approximately 16 percent fewer children than non-Malthusians. Notably, the coefficient on actual fertility is *larger* in absolute value than the coefficient on

⁷All specifications include both age and age squared, so the coefficient on age in the table reflects the linear component only. The marginal effect of age combines linear and quadratic components and is therefore age-specific.

Table 2. Effect of Malthusian Beliefs on Fertility

	Ideal Fertility		Actual Fertility	
	(1)	(2)	(3)	(4)
Malthusian	−0.268*** (0.041)		−0.292*** (0.066)	
Malthusian (continuous)		−0.153*** (0.021)		−0.146*** (0.033)
Age	−0.014** (0.006)	−0.015** (0.006)	0.112*** (0.010)	0.111*** (0.010)
Female	0.015 (0.040)	0.026 (0.040)	0.091 (0.068)	0.104 (0.068)
Education	−0.047*** (0.007)	−0.046*** (0.007)	−0.144*** (0.013)	−0.143*** (0.013)
Year FE	Yes	Yes	Yes	Yes
Observations	1,822	1,822	1,822	1,822
R^2	0.060	0.069	0.262	0.262

Notes: All models include age squared and year fixed effects. Robust standard errors in parentheses. The reported age coefficient is the linear term only. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

ideal fertility, suggesting that beliefs translate into behavior with at least as much force as they shape stated preferences.

3.4 Completed Fertility and Heterogeneity

A concern with actual fertility is that younger respondents may not have completed child-bearing. Table 3 restricts the sample to women aged 45+ and men aged 50+. Among those with completed fertility, Malthusians have 0.26 fewer children than non-Malthusians, similar to the full-sample estimate. The relationship also appears among those who have not completed fertility, suggesting belief-driven differentials emerge early in the reproductive life course.

Table 4 examines heterogeneity by education. The belief–fertility relationship is present across education groups for both outcome measures. For ideal fertility, the coefficient is somewhat larger for less educated respondents; for actual fertility, the pattern reverses, with the coefficient larger among the more educated. This is suggestive rather than dispositive, since I do not test the cross-group difference formally. With that caveat, the descriptive pattern is consistent with the interpretation that Malthusian beliefs affect fertility preferences across the education distribution but translate into behavior more strongly among the more

Table 3. Actual Fertility by Completed Fertility Status

	Full Sample (1)	Completed Fertility (2)	Not Yet Completed (3)
Malthusian	−0.292*** (0.066)	−0.257** (0.109)	−0.288*** (0.079)
Observations	1,822	848	974
R^2	0.262	0.112	0.266

Notes: Dependent variable is actual number of children. All models control for age, age squared, sex, education, and year. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

educated, perhaps because they face fewer constraints on achieving their fertility targets.

Table 4. Heterogeneity in the Belief–Fertility Relationship

	Ideal Fertility		Actual Fertility	
	HS or Less	Some College+	HS or Less	Some College+
Malthusian	−0.318*** (0.069)	−0.235*** (0.049)	−0.209* (0.115)	−0.392*** (0.081)
Observations	821	1,001	821	1,001
R^2	0.045	0.029	0.163	0.240

Notes: Dependent variable is ideal (columns 1–2) or actual (columns 3–4) number of children. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.5 Interpretation and Limitations

The empirical findings are consistent with the model’s central prediction: individuals who believe population growth is unsustainable both desire and have fewer children. The relationships are economically meaningful—gaps of 11 and 16 percent of the respective sample means—and statistically robust.

Two features of the results deserve emphasis. First, the relationship holds for *actual* fertility, not just stated preferences. This addresses a common concern that surveys capture cheap talk rather than consequential behavior. Second, the coefficient on actual fertility is at least as large as the coefficient on ideal fertility, suggesting either that beliefs operate partly through channels beyond conscious intention or that individuals with stronger beliefs are more committed to acting on them.

Several limitations are worth flagging. The evidence remains correlational and could reflect reverse causality or omitted-variable bias. The cross-sectional nature of the data precludes tracking how belief changes affect subsequent fertility decisions. The belief measure also captures concern about global population sustainability, not specifically the scale-effect parameter θ . Despite these limitations, the evidence provides suggestive support for the belief channel emphasized in the model. With this empirical foundation in place, I now turn to the strategic implications.

4 Strategic Interaction and Equilibrium Selection

This section analyzes strategic interaction in fertility choices and partitions the equilibrium space into three economically distinct regions. The analysis yields three results that organize the rest of the paper. First, Romerians exhibit strategic complementarity while Malthusians exhibit strategic substitutability, so the slope of the aggregate best response is a continuous function of the Romerian share that switches sign at an interior threshold. Second, two thresholds ϕ^* and ϕ^{**} partition the share space into a Malthusian-dominated region of unconditional uniqueness, a contraction region of conditional uniqueness, and a Romerian-dominated multiplicity region whose presence depends on the strength of scale effects. Third, in the multiplicity region a global-games refinement à la [Frankel et al. \(2003\)](#) selects a risk-dominant equilibrium, restoring a unique selection across the entire share space.

4.1 The Fertility Game

Consider fertility choice as a game among households. Each household i chooses n_i to maximize expected utility, and aggregate fertility is $\bar{n} = \int n_i di$. The household's payoff depends on $(n_i, \bar{n})$ through a single channel: higher $\bar{n}$ raises $N' = \bar{n}N$, which through (3) affects A'' and subsequent productivity. Let $U_j(n_j, \bar{n})$ denote the expected utility of a type- j household choosing n_j when aggregate fertility is $\bar{n}$, with the expectation taken under belief θ_j . Type j exhibits *strategic complementarity* if $\partial^2 U_j / \partial n_j \partial \bar{n} > 0$ and *strategic substitutability* if the cross-partial is negative.

4.2 Best Responses and the Sign of Strategic Interaction

Let $n_j(\bar{n}^e)$ denote the fertility choice of type j when expected aggregate fertility is $\bar{n}^e$. The actual aggregate is $\bar{n} = \phi n_R(\bar{n}^e) + (1 - \phi) n_M(\bar{n}^e)$, and rational expectations require $\bar{n}^e = \bar{n}$. Each household knows its own type but is uncertain about others' realizations and hence about aggregate fertility.

Under the local log-linear approximation, the marginal effect of expected aggregate fertility on the continuation value is $\partial \mathbb{E}_j[V_j]/\partial \bar{n}^e = \xi_{N,j}/\bar{n}^e$. From (17), the sign of $\xi_{N,j}$ matches the sign of θ_j : $\xi_{N,M} < 0$ for Malthusians and $\xi_{N,R} > 0$ for Romerians. Since the household FOC has a strictly increasing left-hand side in n_j , the sign of $dn_j^*/d\bar{n}^e$ coincides with the sign of $\xi_{N,j}$. The sign result is robust beyond the local approximation, since $\xi_{N,j}$ inherits the sign of θ_j under any smooth value function.

Proposition 1 (Strategic Interaction). *Under Assumptions 1–3: (i) Romerians exhibit strategic complementarity, $dn_R^*/d\bar{n}^e > 0$; (ii) Malthusians exhibit strategic substitutability, $dn_M^*/d\bar{n}^e < 0$. Substituting (17) into the FOC and totally differentiating gives the closed-form local slopes*

$$d_j \equiv \left. \frac{dn_j^*}{d\bar{n}^e} \right|_{ref} = \frac{\beta^2 n_j^* \theta_j}{\gamma \bar{n} (1 - \beta n_j^*)^2}, \quad (18)$$

evaluated at the reference $\bar{n}$ and the equilibrium n_j^ .*

The proof is in Appendix A. The intuition is straightforward. A Romerian who expects higher aggregate fertility expects higher future population, which under Romerian beliefs implies faster productivity growth and a higher continuation value of each child; the Romerian responds by having more children. A Malthusian who expects higher aggregate fertility expects slower future productivity growth, which lowers the continuation value and reduces fertility.

4.3 Three Equilibrium Regions

Define the aggregate best-response slope

$$\rho(\phi) \equiv \phi d_R + (1 - \phi) d_M = \phi d_R - (1 - \phi) |d_M|, \quad (19)$$

with $d_R > 0 > d_M$. As a continuous function of ϕ , $\rho(\phi)$ crosses zero at the threshold

$$\phi^* = \frac{|d_M|}{d_R + |d_M|}, \quad (20)$$

and crosses one at

$$\phi^{**} = \frac{1 + |d_M|}{d_R + |d_M|}, \quad (21)$$

provided the right-hand side is below one—which holds if and only if $d_R > 1$. The two thresholds carve the unit interval into three economically distinct regions, characterized in the following theorem.

Theorem 1 (Equilibrium Existence and Uniqueness by Region). *Suppose Assumptions 1–3 hold and Proposition 1 applies at the relevant reference. Then:*

- (i) **Malthusian-dominated region**, $\phi < \phi^*$. *The aggregate best response $BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1 - \phi)n_M^*(\bar{n}^e)$ is strictly decreasing in a neighborhood of any candidate fixed point. The equilibrium is locally unique unconditionally on the magnitude of $\rho(\phi)$.*
- (ii) **Contraction region**, $\phi^* \leq \phi \leq \phi^{**}$ (or all $\phi \geq \phi^*$ when $d_R \leq 1$). *The aggregate best response satisfies $0 \leq \rho(\phi) < 1$, the contraction-mapping argument applies, and the equilibrium is locally unique.*
- (iii) **Multiplicity region**, $\phi > \phi^{**}$ (only when $d_R > 1$). *The aggregate best response satisfies $\rho(\phi) > 1$ and can intersect the 45-degree line transversally at multiple points, generating an odd number of locally ordered equilibria of which the extreme two are stable.*

The proof is in Appendix A. The economic content is sharp. In the Malthusian-dominated region, optimist coordination is too thin to generate the strategic complementarity needed for multiple stable points, so equilibrium is unconditionally pinned down by the downward-sloping best response. In the contraction region, complementarity is present but bounded, so the standard contraction argument delivers uniqueness. Only in the Romerian-dominated region with sufficiently strong scale effects does complementarity dominate enough to generate multiplicity, and there the question becomes which of the multiple stable equilibria the economy plays. I address that question in the next subsection.

When does the multiplicity region exist? Theorem 1(iii) requires $d_R > 1$, which substituting (18) reads

$$\theta_R > \theta_R^{crit} \equiv \frac{\gamma \bar{n} (1 - \beta n_R^*)^2}{\beta^2 n_R^*}. \quad (22)$$

At the calibration $(\beta, \gamma) = (0.30, 0.325)$ with $n_R^* = 1.2$ and $\bar{n} \approx 1$, $\theta_R^{crit} \approx 1.23$. Empirical estimates of θ_R span a wide range. Pure semi-endogenous scale effects à la Jones (1995); Bloom et al. (2020) place it in $[0.1, 0.3]$; broader notions including agglomeration, idea-recombination effects, and the long-run estimates of Kremer (1993) can reach $\theta_R \in [0.5, 1.5]$; and the unified-growth literature (Galor, 2011) along with the long-run quantitative work it builds on can imply effective θ_R at or above the threshold. *Under standard semi-endogenous-growth calibrations the multiplicity region is empty, while under Kremer–Galor–Weil calibrations it is occupied at high ϕ .* Both cases are interesting, and the analysis that follows accommodates them.

4.4 Stability and Global-Games Selection

Within the multiplicity region the local-stability properties of equilibria provide a first-pass refinement. Linearizing the best-response dynamics around an equilibrium, an equilibrium is stable if and only if $|\rho(\phi)| < 1$ at the slopes evaluated locally at that equilibrium. In the multiplicity region, the slopes vary across the multiple fixed points, and standard transversality arguments imply that consecutive intersections of $BR(\bar{n}^e)$ with the 45-degree line alternate between stable and unstable. With $\rho(\phi) > 1$ on average, the typical configuration is two stable equilibria flanking an unstable middle equilibrium. Stability eliminates the middle equilibrium but leaves the two stable selections to be coordinated upon.

To break this remaining indeterminacy, I adopt the global-games refinement of [Frankel et al. \(2003\)](#), which extends the foundational arguments of [Carlsson and Van Damme \(1993\)](#) and [Morris and Shin \(1998\)](#) to general supermodular games with continuous actions. The key technical step in my setting is a reduction: because Malthusians have a strictly downward-sloping best response and a unique fixed point given any aggregate fertility level, their best response can be substituted out. The reduced game among Romerians is then supermodular and falls within the scope of [Frankel et al. \(2003\)](#).

Theorem 2 (Global-Games Selection in the Multiplicity Region). *Consider the multiplicity region of Theorem 1(iii). Suppose households observe private signals $\phi_i = \phi + \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, \sigma_\zeta^2)$ and the prior on ϕ is diffuse. As $\sigma_\zeta \rightarrow 0$, iterated deletion of strictly dominated strategies in the reduced Romerian subgame selects a unique threshold equilibrium with cutoff $\phi^{rd}(\theta_R, \theta_M, \beta, \gamma) \in (\phi^{**}, 1)$ at which Romerians coordinate on the high-fertility selection above the cutoff and on the low-fertility selection below. The selected equilibrium is risk-dominant in the sense of [Frankel et al. \(2003\)](#).*

The proof, including the construction of the reduced subgame, the verification of the supermodularity and monotonicity conditions of [Frankel et al. \(2003\)](#), and the closed-form indifference equation for ϕ^{rd} , is in Appendix A. The result restores a unique equilibrium selection across the entire ϕ -space: Theorem 1 delivers uniqueness in the first two regions, and Theorem 2 delivers selection in the third. The cutoff ϕ^{rd} is a closed-form function of preferences and beliefs, and it is the empirical object that the planner’s announcement targets in the multiplicity case.

Why selection matters for policy. Outside the multiplicity region, an announcement plays a single role: it shifts households’ subjective posteriors over θ , and through them the unique equilibrium fertility. Inside the multiplicity region, an announcement plays a second role: by shifting the prior over ϕ (or, equivalently, by signaling the intended selection), it can

move the economy across the cutoff ϕ^{rd} and switch the selection from low to high (or vice versa). The selection-shifting role is most relevant when the prior ϕ is near ϕ^{rd} and absent when the prior is far from ϕ^{rd} on either side. Section 7 treats both roles.

4.5 Map of Regions and Empirical Relevance

Table 5 summarizes the partition. The trap region of Section 5 typically features ϕ below the critical share at which aggregate fertility just reaches replacement, which is itself well below ϕ^{**} for any standard calibration; the trap therefore sits inside the contraction (or even Malthusian-dominated) region, and the policy intuition for the trap is unaffected by multiplicity. The interior steady state of the Malthusian world studied in Section 5.3 also sits in the contraction region for standard calibrations. Multiplicity, when present, is most likely in modern Romerian-dominated economies with strong scale effects, where the announcement plays its most distinctive role.

Table 5. Equilibrium Regions and Their Economic Content

Region	Condition	$\rho(\phi)$	Equilibrium	Where it applies
Malthusian-dominated	$\phi < \phi^*$	< 0	Unique unconditionally	Trap; pre-industrial era
Contraction	$\phi^* \leq \phi \leq \phi^{**}$	$[0, 1)$	Unique via contraction	Most calibrations
Multiplicity	$\phi > \phi^{**}$	> 1	Multiple, selection by GG	Strong scale effects

Notes: Multiplicity region is non-empty if and only if $\theta_R > \theta_R^{crit}$, with $\theta_R^{crit} \approx 1.23$ at the baseline calibration $(\beta, \gamma) = (0.30, 0.325)$ and $n_R^* = 1.2$. GG denotes the global-games selection of Theorem 2.

The stabilizing role of Malthusians. An additional feature distinguishes the present setting from pure-complementarity coordination games. Because Malthusian best responses slope downward, the Malthusian fraction $1 - \phi$ acts as a continuous shifter that pushes $\rho(\phi)$ down. Compared to a population of only Romerians, the heterogeneous mix delivers a larger contraction region and a smaller multiplicity region. Pessimists, paradoxically, stabilize the equilibrium structure of the economy.

5 Long-Run Dynamics and the Role of Wrong Beliefs

I now analyze the long-run dynamics of beliefs and population. Wrong beliefs generate qualitatively different pathologies depending on which worldview is correct. In Romerian worlds, Malthusian minorities cause prolonged stagnation through an informational poverty trap. In Malthusian worlds, Romerian minorities generate non-monotone dynamics, including the

boom-bust cycles familiar from pre-industrial demography. The asymmetry arises from the interaction between demographic selection (which always favors optimists) and Bayesian learning (which favors correct beliefs, but only when signals are informative).

5.1 Two Forces: Demographic Selection and Learning

The Romerian share ϕ_t evolves according to (13), driven by two forces.

Demographic selection. Since $n_R > n_M$, $\phi_t^{birth} > \phi_t$ for any interior ϕ_t . Demographic selection pushes the cultural-transmission part of ϕ_{t+1} upward regardless of which belief is correct.

Bayesian learning. The direction of π_{t+1} depends on the true θ^* . If $\theta^* > 0$, high population tends to generate high productivity growth—evidence for the Romerian model; if $\theta^* < 0$, high population tends to generate low productivity growth—evidence for the Malthusian model. The *strength* of learning depends on population through the signal informativeness $\mathcal{I}_t$ in (12). When $N_t \approx \bar{N}$, $\mathcal{I}_t \approx 0$, and learning is weak regardless of the truth.

The interaction between these two forces produces sharply different dynamics in Romerian and Malthusian worlds. In Romerian worlds, both forces align toward correct beliefs; the only question is how quickly correct beliefs spread, and the answer can be “arbitrarily slowly” when the economy starts in the trap. In Malthusian worlds, the two forces conflict, and the resulting tension generates the non-monotone dynamics that the rest of this section makes precise.

5.2 Dynamics in a Romerian World: Convergence and the Trap

Theorem 3 (Convergence in Romerian Worlds). *Suppose $\theta^* > 0$ and that the population trajectory remains within the region where (3) is well-defined. Starting from any interior initial condition $(\phi_0, N_0, A_0, \pi_0)$: (i) $\pi_t \rightarrow 1$ almost surely; (ii) $\phi_t \rightarrow 1$ almost surely; (iii) aggregate fertility converges to n_R^* ; (iv) $\sup_t \ln(N_t/\bar{N}) = +\infty$ almost surely.*

The proof is in Appendix A. Both demographic selection and learning favor Romerians when the world is truly Romerian: selection favors them because they have more children, and learning favors them because observed outcomes confirm their beliefs.⁸

⁸The result that $\sup_t \ln(N_t/\bar{N}) = +\infty$ presumes the absence of endogenous bounds on population. The framework can be extended to deliver bounded population in two natural ways: a hard carrying capacity above which fertility returns to replacement (modeled as a constraint on ψ), or endogenous congestion through which the effective time cost rises with population density. Under either extension, $\phi_t \rightarrow 1$ almost surely, but population converges to a stochastic neighborhood of the carrying capacity rather than diverging.

Theorem 3 guarantees almost-sure convergence but is silent on the rate. The next analysis shows that convergence can be arbitrarily slow when the economy starts in the trap region.

The Uninformative Region.

Definition 2 (Uninformative Region). *For $\delta > 0$, the δ -uninformative region is $\{N_t : |\ln(N_t/\bar{N})| < \delta\}$. In this region, the per-period information bound is $\mathcal{I}_t < (\theta_R - \theta_M)^2 \delta^2 / (2\sigma_\eta^2) \equiv \bar{\mathcal{I}}(\delta)$.*

When δ is small, $\bar{\mathcal{I}}(\delta)$ is small, and signals reveal almost nothing about the true θ^* .

Definition 3 (Trap Region). *The informational poverty trap region $\mathcal{T}(\delta, \zeta)$ is the set of states (ϕ, N, π) with: (i) $|\ln(N/\bar{N})| < \delta$; (ii) $\bar{n}(\phi) \in [1, 1 + \zeta]$ for small $\zeta > 0$; and (iii) $\pi \leq \phi$, so posterior pessimism aligns with prevalent Malthusianism.*

The third condition ensures the trap is closed under the joint dynamics of (ϕ_t, N_t, π_t) : once $\pi_t \leq \phi_t$, this inequality propagates inductively (see proof of Proposition 2).

Assumption 4 (Fertility Straddles Replacement). *Malthusian fertility is below replacement and Romerian fertility above: $n_M < 1 < n_R$.*

Under Assumption 4, there is a critical Romerian share $\phi^{trap} = (1 - n_M)/(n_R - n_M)$ at which $\bar{n} = 1$. For ϕ slightly above ϕ^{trap} , aggregate fertility slightly exceeds 1 and population grows slowly. Note that ϕ^{trap} is well within the Malthusian-dominated or contraction regions of Theorem 1 for standard calibrations, so the trap analysis is unaffected by any multiplicity issues elsewhere in the share space.

Proposition 2 (Slow Learning in the Trap). *Consider a Romerian world ($\theta^* = \theta_R > 0$) under Assumption 4. Suppose $(\phi_0, N_0, \pi_0) \in \mathcal{T}(\delta, \zeta)$ for small $\delta, \zeta > 0$. For target $\bar{\phi} > \phi_0$, define the escape time $T^{esc} = \inf\{t : \phi_t > \bar{\phi}\}$. Then, as long as the trajectory remains in $\mathcal{T}(\delta, \zeta)$:*

- (i) $|\mathbb{E}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t]| \leq C_1 \bar{\mathcal{I}}(\delta)$ with $C_1 = 1/4$;
- (ii) $\phi_{t+1}^{birth} - \phi_t \leq C_2(n_R - n_M)(1 - \phi_t)$ with $C_2 = 1/n_M$;
- (iii) $\mathbb{E}[T^{esc}] \geq (\bar{\phi} - \phi_0)/D$, where $D = \lambda C_2(n_R - n_M) + (1 - \lambda)C_1 \bar{\mathcal{I}}(\delta)$.

The informational poverty trap is unaffected because it operates at populations close to $\bar{N}$, far from any carrying capacity.

The proof is in Appendix A. As $\delta \rightarrow 0$, the learning contribution vanishes and escape relies on demographic selection, which is itself slow when $n_R - n_M$ or λ is small. Corollary 3 in the appendix shows that the expected escape time diverges as $\Omega(1/\delta^2)$ when learning is the binding channel and can exceed any finite horizon.

Proposition 7 in the appendix characterizes the boundary of the trap. The critical population beyond which learning dominates demographic selection is

$$N^{crit}(\phi)/\bar{N} = \exp\left(\sqrt{\frac{2\sigma_\eta^2 \cdot \lambda C_2(n_R - n_M)(1 - \phi)}{(1 - \lambda)C_1(\theta_R - \theta_M)^2}}\right).$$

High noise σ_η^2 raises N^{crit} , making the trap larger; strong cultural transmission λ also raises N^{crit} by reducing reliance on learning; and large belief dispersion $(\theta_R - \theta_M)^2$ lowers N^{crit} by making signals more diagnostic when population deviates from $\bar{N}$.

In a Romerian world, then, Malthusian minorities cause prolonged stagnation through a self-reinforcing mechanism: pessimism suppresses fertility, low fertility keeps population near $\bar{N}$ (the uninformative region), and uninformative signals fail to correct pessimism. Escape eventually occurs but can take arbitrarily long.

5.3 Dynamics in a Malthusian World: Opposing Forces and Non-Monotone Outcomes

The dynamics differ qualitatively when $\theta^* < 0$. Whereas in a Romerian world demographic selection and learning push the Romerian share ϕ_t in the same direction, in a Malthusian world they pull in opposite directions, with the balance between them itself depending on the state.

Proposition 3 (Opposing Forces in Malthusian Worlds). *Suppose $\theta^* = \theta_M < 0$. At any interior state with $\phi_t \in (0, 1)$ and $\pi_t \in (0, 1)$:*

- (i) Demographic-selection drift on births is strictly positive and bounded. *The contribution of differential fertility to the share of Romerian-parented children satisfies*

$$0 < \phi_t^{birth} - \phi_t = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t} \leq \frac{n_R - n_M}{4n_M},$$

independently of $\ln(N_t/\bar{N})$.

- (ii) Bayesian learning drift on the posterior is strictly negative under truth and grows quadratically in population deviation. *The expected change in the posterior under the*

true law of motion is

$$\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t] = -\pi_t(1 - \pi_t)\mathcal{I}_t + o(\mathcal{I}_t), \quad \mathcal{I}_t = \frac{(\theta_R - \theta_M)^2 [\ln(N_t/\bar{N})]^2}{2\sigma_\eta^2}.$$

(iii) The expected drift in ϕ_t admits a sign change as a function of population deviation.

Using (13), the joint drift decomposes as

$$\mathbb{E}^{\theta_M}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] = \underbrace{\lambda(\phi_t^{birth} - \phi_t)}_{\geq 0, \text{ selection}} + \underbrace{(1 - \lambda)(\pi_t - \phi_t)}_{\text{cultural regression toward } \pi_t} + \underbrace{(1 - \lambda)\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t]}_{\leq 0, \text{ learning}}.$$

At $\ln(N_t/\bar{N}) = 0$ the learning term vanishes, so the drift equals $\lambda(\phi_t^{birth} - \phi_t) + (1 - \lambda)(\pi_t - \phi_t)$; whenever $\pi_t \geq \phi_t - \frac{\lambda}{1-\lambda}(\phi_t^{birth} - \phi_t)$, this is non-negative. For $|\ln(N_t/\bar{N})|$ sufficiently large, the learning term dominates and the drift is strictly negative whenever $\pi_t \in (0, 1)$. The threshold $\bar{L}(\phi_t, \pi_t, \lambda)$ at which the sign switches is given explicitly in the proof. In particular, the deterministic skeleton admits no monotone law of motion in ϕ_t .

The proof, in Appendix A, is a direct Bayes log-odds calculation combined with a bounded-selection-drift comparison. The threshold takes the form

$$\bar{L}(\phi_t, \pi_t, \lambda)^2 = \frac{2\sigma_\eta^2 \cdot [\lambda(\phi_t^{birth} - \phi_t) + (1 - \lambda)\max\{\pi_t - \phi_t, 0\}]}{(1 - \lambda)\pi_t(1 - \pi_t)(\theta_R - \theta_M)^2/1},$$

which is decreasing in the Romerian–Malthusian fertility gap and increasing in signal noise.

Economic content. Proposition 3 captures the core economic asymmetry between the two worlds. In a Romerian world (Theorem 3), demographic selection and learning both favor Romerians, and trajectories converge monotonically toward $\phi = 1$ almost surely. In a Malthusian world, the two forces conflict. Demographic selection always pushes upward because optimists outbreed pessimists; learning under truth always pushes downward because outcomes confirm the Malthusian model; and the magnitude of the learning channel scales quadratically with how far population has deviated from the reference. The drift is therefore non-negative near $\bar{N}$ (where signals are uninformative and demographic selection dominates the cultural-regression term whenever π_t is not too small) and strictly negative once population has grown enough that the learning channel overtakes selection. The system cannot settle by drifting monotonically in either direction. Any candidate interior steady state must lie at a population strictly above the reference, where the deterministic component of log-productivity growth vanishes: $\bar{g} + \theta_M \ln(N^{ss}/\bar{N}) = 0$ gives $\ln(N^{ss}/\bar{N}) = -\bar{g}/\theta_M > 0$.

Cycle dynamics. The state-dependent reversal in Proposition 3 naturally produces the cyclical pattern familiar from economic-historical accounts of the pre-industrial era, with three identifiable phases. *Expansion phase:* population is moderate, $|\ln(N_t/\bar{N})|$ is small, learning is weak, demographic selection dominates, and ϕ_t rises. *Crisis phase:* as ϕ_t and population grow, signals become informative; productivity slows because $\theta^* < 0$; the Malthusian model fits much better than the Romerian; π_{t+1} falls sharply, overwhelming demographic selection; and ϕ_{t+1} falls. *Recovery phase:* fertility falls, population stabilizes or contracts, signals weaken, demographic selection regains the upper hand, and the pattern can repeat. Whether the system traces out a deterministic limit cycle, damped oscillations around the interior steady state, or stochastic recurrence around it depends on the curvature of the nonlinear dynamics and the size of the structural shocks. What Proposition 3 establishes is that all of these belong to the family of non-monotone dynamics the system can produce, and that none can collapse to monotone convergence.

Comparative statics for the strength of cycles. The threshold $\bar{L}$ and the underlying drift magnitudes deliver clean comparative statics for how pronounced the non-monotone dynamics are. When learning is very weak (very high σ_η^2 or very small belief gap $\theta_R - \theta_M$), $\bar{L}$ is large and the drift remains positive over a wide population range; demographic selection dominates and ϕ_t rises persistently, so the economy lingers in an inefficient high-population, optimist-dominated configuration with low welfare. When learning is very strong (very low σ_η^2), $\bar{L}$ is small and the drift turns negative for modest population deviations; Malthusian beliefs spread quickly, ϕ_t falls toward zero, and the system settles into a Malthusian-dominated regime. The most pronounced non-monotone dynamics—the boom-bust cycles with both phases visible—arise for intermediate learning responsiveness, where neither force dominates globally and the drift sign reversal repeatedly engages.

5.4 The Asymmetry: Why Wrong Beliefs Generate Different Pathologies

Table 6 summarizes the asymmetry.

In Romerian worlds, both forces align toward correct beliefs. Incorrect (Malthusian) beliefs are self-correcting, but the correction can be very slow if the economy starts in the trap. In Malthusian worlds, the forces conflict: demographic selection favors optimists who happen to be wrong, while learning favors the correct pessimistic model. Proposition 3 shows that this conflict produces a state-dependent drift that cannot be resolved by monotone convergence, opening the door to the non-monotone dynamics that economic historians have documented.

This asymmetry has a profound implication. Human societies are more vulnerable to

Table 6. Pathologies from Wrong Beliefs

	Romerian World ($\theta^* > 0$)	Malthusian World ($\theta^* < 0$)
Wrong belief	Malthusian ($\theta_M < 0$)	Romerian ($\theta_R > 0$)
Demographic selection	Favors Romerians (correct)	Favors Romerians (incorrect)
Learning	Favors Romerians (correct)	Favors Malthusians (correct)
Forces align?	Yes	No
Pathology	Informational poverty trap	Non-monotone dynamics, including cycles
Long-run outcome	Convergence to truth	State-dependent reversal

excessive optimism in terms of stability, but more vulnerable to excessive pessimism in terms of stagnation. Optimists reproduce more regardless of whether their optimism is warranted, which destabilizes Malthusian worlds. Pessimism keeps population near the reference and thereby suppresses learning, which traps Romerian worlds.

Remark 1 (The Core Insight). *The central theoretical contribution can be stated sharply: in a Romerian world, Malthusians cause stagnation; in a Malthusian world, Romerians generate non-monotone dynamics, including cycles. Wrong beliefs are always costly, but the nature of the cost depends on the direction of the error.*

6 Demographic History: Two Interpretations

The model offers a unified framework for interpreting broad patterns in demographic history. It admits two fundamentally different interpretations of the pre-industrial era, with distinct implications for understanding the modern world.

6.1 Interpretation 1: A Genuinely Malthusian Past

The conventional interpretation, consistent with economic historians from Malthus himself to [Clark \(2007\)](#), is that the pre-industrial world was genuinely Malthusian: scale effects were negative, and larger populations did reduce living standards.

Under this interpretation, the cycles documented by [Voigtländer and Voth \(2013\)](#) and others—periods of demographic expansion followed by crises that reduced population—reflect exactly the kind of non-monotone dynamics established in Proposition 3. During periods of relative prosperity, Romerian-like optimism spread through demographic selection; population grew; eventually, population deviation from the reference became large enough that the learning channel engaged; living standards fell, Malthusian beliefs were reinforced by crisis, and the drift in ϕ_t reversed. Population stabilized or declined, recovery began, and the pattern restarted.

The Black Death of 1347–1351 provides a particularly clear example. Europe’s population had grown substantially in the preceding centuries, straining resources. The plague killed roughly one-third of the population. Real wages rose sharply in the aftermath and remained elevated for over a century. The model interprets this as a crisis phase that both directly reduced population and reinforced Malthusian beliefs, leading to a sustained period of lower fertility.

Under this interpretation, the Industrial Revolution marked a genuine regime change from $\theta^* < 0$ to $\theta^* > 0$. Sustained technological progress, enabled by the accumulation of scientific knowledge and the development of institutions supporting innovation, broke the Malthusian link between population and living standards. The world shifted from being truly Malthusian to truly Romerian.

6.2 Interpretation 2: An Always-Romerian World Trapped by Information Poverty

The informational poverty trap opens a radically different interpretation: the world may have *always* been Romerian, but humanity was trapped for millennia by the self-reinforcing dynamics of pessimism and a population stuck near the reference level.

Under this interpretation, positive scale effects existed throughout history. More people always meant more ideas, more specialization, more innovation. But populations were too close to the relevant reference, and productivity signals too noisy, for anyone to reliably learn this fact. Malthusian beliefs, once established, suppressed fertility, keeping population near $\bar{N}$. A population in this region generated uninformative signals, and uninformative signals failed to correct Malthusian beliefs. The trap persisted not because Malthusianism was correct, but because the evidence needed to refute it could only be generated by escaping it.

The observed pre-industrial cycles, under this interpretation, were not the inevitable boom-bust dynamics of a Malthusian world but rather failed escape attempts from the trap. Occasionally, some region or era experienced conditions that pushed population higher: better weather, new crops, reduced warfare. Beliefs began to update toward optimism. But the gains were fragile. Any negative shock—plague, famine, invasion—was interpreted as confirmation of Malthusian fears, beliefs snapped back, fertility fell, and the economy returned to the trap.

The Industrial Revolution, under this interpretation, was not a technological regime change but an *informational escape*. Some combination of factors—the Enlightenment, improving institutions, a sequence of positive productivity shocks—pushed European populations above the critical threshold N^{crit} . Once in the informative region, learning accelerated;

evidence of positive scale effects accumulated; beliefs shifted decisively toward Romerianism; fertility rose further; population exploded. The trap dissolved, not because the world changed, but because humanity finally accumulated enough information to see what had always been true.

6.3 Distinguishing the Interpretations

Both interpretations are consistent with the qualitative features of demographic history: long periods of stagnation, occasional expansions, crises, and eventual escape to modern growth. Distinguishing them requires careful attention to details of the historical record and to quantitative magnitudes.

Several features favor the genuine-regime-change interpretation. First, the timing of the Industrial Revolution coincides with specific technological innovations (steam power, mechanization) that plausibly altered the relationship between population and productivity. Second, the scale of technological progress after 1800 suggests a qualitative break rather than just informational updating. Third, environmental constraints (land, resources) appear to have genuinely bound pre-industrial economies in ways that seem inconsistent with large positive scale effects.

Several features favor the informational-escape interpretation. First, there is evidence of positive scale effects even in pre-industrial settings: larger cities were more innovative, and more connected regions were more prosperous. Second, the speed of the transition, once it began, is more consistent with belief updating than with gradual technological change. Third, the modern fertility decline in developed countries (discussed below) is puzzling if the world is truly Romerian, but explicable if beliefs can drift back toward Malthusianism even in a Romerian world.

I do not take a definitive stand on which interpretation is correct. The contribution of the model is to show that *both are theoretically coherent* and that they have *different implications for policy*.

6.4 The Modern Fertility Decline

Fertility rates in developed countries have fallen dramatically since the 1960s, reaching levels well below replacement in most of Europe and East Asia. This decline is puzzling from a simple Romerian perspective: if larger populations are beneficial, why are the most prosperous societies choosing the fewest children? The model suggests several interpretations.

Revival of Malthusian beliefs. The environmental movement, which emerged prominently in the 1960s and 1970s, can be understood as a revival of Malthusian thinking. Concerns about pollution, resource depletion, climate change, and ecological limits represent beliefs that population growth is harmful. The empirical evidence in Section 3 documents that such beliefs are prevalent and are associated with substantially lower fertility. Under the informational-escape interpretation, the modern fertility decline represents a partial return toward the trap.

Actual shift to Malthusian technology. Under the regime-change interpretation, the true θ^* may have recently shifted from positive to negative. Automation and AI may have reduced the marginal contribution of human labor to economic output. Environmental constraints, particularly climate change, may have imposed genuine costs on population growth.

Quantity-quality substitution. The model abstracts from human capital, but in reality parents face a quantity-quality tradeoff. Rising returns to education may have induced parents to invest more in each child’s human capital at the expense of having more children. This is conceptually distinct from Malthusianism but generates similar fertility outcomes.

Under the dynastic welfare criterion of Section 7, what matters is whether households’ beliefs match the truth: information policies that correct misperceptions are warranted only if beliefs are wrong. Under the total utilitarian criterion, the planner additionally values population for its own sake, so even a correct-belief decentralized equilibrium leaves room for intervention through fertility subsidies or optimism-promoting communication.

6.5 Contemporary Divergence and Future Dynamics

Fertility today varies enormously across countries and groups. Total fertility rates range from below 1.0 in some East Asian countries to above 6.0 in parts of Sub-Saharan Africa. Within countries, fertility varies systematically with religion, education, and political orientation.

The model predicts that these differences are partially driven by belief heterogeneity and are self-reinforcing through demographic selection. Groups with Romerian beliefs (or with religious or cultural traditions that encourage high fertility for other reasons) will grow in population share over time. If cultural transmission is strong, these compositional shifts can substantially affect aggregate beliefs and fertility over generations. Standard demographic projections typically assume convergence to low fertility as countries develop, but the model suggests convergence may be slower or incomplete if belief heterogeneity persists. Whether

this is efficient depends on the true value of θ^* : if Romerian, high-fertility groups move the population toward efficiency; if Malthusian, they exacerbate the problem.

7 Policy: Welfare Criteria and the Value of Public Information

I now analyze the welfare effects of public announcements about the scale-effect parameter θ . The analysis combines two mechanisms: an information-provision channel that operates throughout the share space, and a selection-shifting channel that operates inside the multiplicity region of Theorem 1. Because standard welfare theorems fail when the set of agents depends on allocations (Goloso et al., 2007), I devote particular care to welfare evaluation.

7.1 Welfare Criteria with Endogenous Population

Evaluating welfare when fertility is a choice variable raises fundamental conceptual issues: the set of people who exist depends on policy, so we cannot simply ask whether everyone is better off because the “everyone” changes. Following Goloso et al. (2007) and the population-ethics literature surveyed by Blackorby et al. (2005), I focus on two normatively distinct criteria.

Dynastic Welfare. Under the dynastic criterion, the planner respects households’ subjective preferences:

$$W^{dyn} = \phi_0 V_R(A_0, N_0, \phi_0, \pi_0) + (1 - \phi_0) V_M(A_0, N_0, \phi_0, \pi_0). \quad (23)$$

This criterion is closely related to the P-efficiency concept of Goloso et al. (2007), which compares allocations using the welfare of the agents who exist in the original allocation.

Total Utilitarian Welfare. The total utilitarian planner sums period utilities across all individuals alive at each date, evaluated under truth:

$$W^{tot} = \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t N_t \cdot \ln c_t \right]. \quad (24)$$

For (24) to be well-defined, $\beta \mathbb{E}^* n_R^* < 1$ at the relevant equilibrium fertility. Under the time-cost specification, $\ln c_t = \ln A_t + \ln(1 - \psi(n_t))$ decomposes additively into a productivity term and a fertility-cost term, making the planner’s tradeoffs transparent.

Average utilitarian benchmark. For comparison, define

$$W^{avg} = \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t \ln c_t \right]. \quad (25)$$

W^{avg} does internalize the per-capita scale-effect channel, since $\ln c_t$ depends on $\ln A_t$, which depends on past population through (4). What W^{avg} omits relative to W^{tot} is the *population-multiplier* term N_t , which gives the total utilitarian planner an additional motive to value larger populations beyond the per-capita welfare effect.

The dynastic and total utilitarian criteria span a substantively distinct normative spectrum. The former treats existing households as the moral universe and respects their preferences; the latter treats all individuals who ever exist as the moral universe and evaluates outcomes under truth. The contrast is the locus of the genuinely interesting policy results: the dynastic planner is fundamentally an information provider, while the total utilitarian planner is an externality corrector. The average-utilitarian benchmark sits between them, internalizing the per-capita externality but not the population motive.

7.2 The First-Best Allocation and Welfare Decomposition

A planner who knows θ^* and can directly choose a symmetric fertility path under the total utilitarian criterion solves

$$\max_{\{n_t\}} \mathbb{E}^* \left[\sum_{t=0}^{\infty} \beta^t N_t \cdot (\ln A_t + \ln(1 - \psi(n_t))) \right] \quad (26)$$

subject to $N_{t+1} = n_t N_t$ and (3). Conjecturing a value function of the local log-linear form $V^P(A, N) = N \cdot w^P(A, N)$ with $w^P(A, N) \approx v^P + \xi_A^P \ln A + \xi_N^P \ln(N/\bar{N})$, the planner's FOC for fertility is

$$\frac{\psi'(n_t^{FB})}{1 - \psi(n_t^{FB})} = \beta [w^P(A', N') + \xi_N^P]. \quad (27)$$

The household's common-correct-belief FOC is structurally identical except for the absence of ξ_N^P . The wedge between planner and household optimality is therefore exactly $\beta \xi_N^P$, with the sign of θ^* .

Proposition 4 (First-Best vs. Decentralized Fertility). *In a Romerian world ($\theta^* > 0$): (i) $n^{FB} > n^{CB}$, where n^{CB} is the common-correct-belief decentralized fertility; (ii) $n^{FB} > n^{CB} > n_M^*$; (iii) the welfare gap decomposes into externality, mean-bias, and dispersion components. In a Malthusian world ($\theta^* < 0$), $n^{FB} < n^{CB}$.*

The proof is in Appendix B. In decentralized equilibrium, households do not internalize

the scale-effect externality. In a Romerian world, this leads to under-investment even with correct beliefs, and belief heterogeneity adds two further sources of inefficiency.

Proposition 5 (Welfare Decomposition). *Under the total utilitarian criterion,*

$$W^{FB}(\theta^*) - W^{HB} = \underbrace{\Delta_1(\theta^*)}_{\text{externality}} + \underbrace{\Delta_2(\theta^*, \bar{\theta})}_{\text{mean bias}} + \underbrace{\Delta_3(\text{Var}(\theta_i))}_{\text{dispersion}}, \quad (28)$$

where $\Delta_1 \geq 0$ with equality iff $\theta^* = 0$; $\Delta_2 \geq 0$ is increasing in $(\bar{\theta} - \theta^*)^2$; and $\Delta_3 \approx (\omega\kappa^2/2) \text{Var}(\theta_i)$.⁹

The proof is in Appendix B. Public announcements can reduce Δ_2 (shifting mean beliefs toward truth) and Δ_3 (reducing dispersion) but cannot directly address Δ_1 without additional policy instruments such as fertility subsidies. A planner with both an information instrument and a fertility instrument could close all three gaps. A planner with only the information instrument can close Δ_2 and Δ_3 directly and, as I show next, use the announcement strategically to compensate *partially* for Δ_1 .

The decomposition delivers three economically distinct messages. First, even with average beliefs perfectly calibrated ($\bar{\theta} = \theta^*$, so $\Delta_2 = 0$), the externality Δ_1 persists: households do not internalize the productivity spillover from their own fertility. Second, even with no dispersion and correct mean beliefs, the externality remains. Third, dispersion generates welfare loss even when $\bar{\theta} = \theta^*$, because households' fertility choices cease to coincide with the symmetric social optimum.

7.3 The Planner's Announcement Problem: Two Channels

Consider a planner who observes a noisy estimate of the true parameter,

$$\hat{\theta} = \theta^* + \nu, \quad \nu \sim \mathcal{N}(0, \sigma_\nu^2), \quad (29)$$

and publicly releases a signal $z = \mu + \tilde{\nu}$, where μ is the chosen announcement value and $\tilde{\nu} \sim \mathcal{N}(0, \tau_A^{-1})$ is the noise households perceive. The announcement operates through two distinct channels.

(1) *Information-provision channel.* Households use z in their Bayes update over the two states $\{\theta_R, \theta_M\}$, shifting the posterior π_t^{post} toward the announced value. This channel operates throughout the share space and is the only channel active outside the multiplicity region of Theorem 1.

⁹The Taylor expansion underlying Δ_3 is taken around the social-optimum fertility n^{SP} , so the externality Δ_1 appears explicitly rather than being absorbed into the dispersion remainder.

(2) *Selection-shifting channel.* In the multiplicity region, the announcement also serves as a public signal about which equilibrium will be played. Through the global-games selection of Theorem 2, an announcement can move the cutoff ϕ^{rd} and switch the selection between the low-fertility and high-fertility stable equilibria. This channel is dormant when the economy is far from the multiplicity boundary and active when it approaches it.

The two channels reinforce each other in Romerian worlds with strong scale effects: an optimism-shaded announcement raises the subjective θ^{post} (information channel), which raises Romerian fertility, which moves ϕ toward the multiplicity region; if the economy enters the multiplicity region, the same announcement helps coordinate on the high-fertility selection (selection channel). They can conflict in Malthusian worlds.

Assumption 5 (Announcement Credibility). *Households treat the announcement as a Gaussian signal with precision $\tau_A > 0$. When $\tau_A \rightarrow \infty$, households fully adopt the announced value; when $\tau_A \rightarrow 0$, households ignore the announcement.*

I take τ_A as exogenous.¹⁰

7.4 Optimal Announcement under Different Criteria

The first set of results characterizes the announcement under each welfare criterion, holding fixed the information-provision channel; the selection-shifting channel is treated separately below.

Theorem 4 (Optimal Announcement: Dynastic Welfare). *Under W^{dyn} , the optimal announcement is truthful in expectation: $\mu^{dyn} = \hat{\theta}$, and the planner prefers maximal informativeness.*

The economic content is sharp. Under the dynastic criterion, the planner is fundamentally an information provider. There is no externality the planner cares about beyond what households themselves would care about, so distorting the announcement strictly reduces welfare.

Theorem 5 (Optimal Announcement: Total Utilitarian Welfare). *Consider $\theta^* = \theta_R > 0$. Under W^{tot} , the optimal announcement satisfies*

$$\mu^{tot} = \hat{\theta} + \chi^{tot}, \quad (30)$$

where $\chi^{tot} > 0$ is an “optimism shade” whenever the externality $\Delta_1 > 0$. The shade has the following comparative statics: (i) χ^{tot} is increasing in $|\theta^*|$; (ii) χ^{tot} is increasing in

¹⁰A microfoundation through repeated-game reputation would make τ_A the equilibrium of a sub-game in which households update on the planner’s history of announcements.

$|\ln(N_0/\bar{N})|$; (iii) χ^{tot} is decreasing in $\text{Var}(\theta_{i,0})$; (iv) χ^{tot} is increasing in announcement credibility τ_A . In a Malthusian world ($\theta^* < 0$), the sign of χ^{tot} reverses: the planner shades toward pessimism.

The proof is in Appendix B. The intuition is that the planner equates the marginal utilitarian benefit of shifting average fertility (carrying an externality wedge of size $\beta\xi_N^P$) to the marginal cost of distorting individual fertility decisions away from their belief-conditional optimum.

Corollary 1 (Optimal Announcement: Average Utilitarian Welfare). *Under W^{avg} , the optimal announcement is $\mu^{avg} = \hat{\theta} + \chi^{avg}$, where $0 < \chi^{avg} < \chi^{tot}$ in a Romerian world (with the inequalities reversing in a Malthusian world). The wedge driving χ^{avg} is the per-capita scale-effect spillover $\beta\xi_{N,avg}^P$, which is strictly smaller in absolute value than the total utilitarian wedge $\beta\xi_N^P$ because it omits the population-multiplier term N_t from (24).*

Corollary 2 (Ordering of Announcements). *In a Romerian world, $\mu^{dyn} = \hat{\theta} < \mu^{avg} < \mu^{tot}$, with the inequalities reversing in a Malthusian world. The total utilitarian planner promotes Romerian beliefs more aggressively than the average utilitarian, who in turn promotes them more aggressively than the dynastic planner.*

7.5 The Selection-Shifting Channel and the Planner's Dilemma

When the economy is in or near the multiplicity region of Theorem 1, the announcement operates additionally through the selection channel. The cutoff ϕ^{rd} of Theorem 2 is a function of expected fertility levels under each selection, both of which respond to households' subjective θ^{post} . An optimism-shaded announcement therefore not only raises Romerian fertility on the high-fertility selection (information channel) but also lowers the cutoff ϕ^{rd} at which the high-fertility selection is played (selection channel). The two channels operate in the same direction in Romerian worlds and reinforce each other in the multiplicity region.

Proposition 6 (Selection-Shifting Effect). *In the multiplicity region, the optimal total utilitarian announcement satisfies $\mu^{tot,sel} = \mu^{tot} + \chi^{sel}$, where $\chi^{sel} > 0$ in a Romerian world reflects the additional benefit from shifting the cutoff ϕ^{rd} toward lower values. The magnitude of χ^{sel} is non-monotone in ϕ : it peaks when ϕ is near ϕ^{rd} and vanishes when ϕ is far from it on either side.*

The proof is in Appendix B. The non-monotonicity is intuitive: when the economy is firmly in the high-fertility basin, shifting the cutoff has no effect on the realized selection; when the economy is firmly in the low-fertility basin, the cutoff is far from binding; selection-shifting matters only at the margin between the two.

A key insight emerges from the structure of the learning problem: the planner faces the same informational constraints as households. If the economy is in the uninformative region, the historical productivity signals that inform $\hat{\theta}$ are themselves uninformative. The planner's signal precision is endogenously low precisely when the trap is most binding.

Remark 2 (The Planner's Dilemma). *Public information is most valuable in the informational poverty trap, where coordinating beliefs could push population above N^{crit} and initiate escape. But this is precisely when the planner has the least information. The cases where intervention matters most are the cases where the planner is most uncertain about what to say.*

This creates a policy dilemma distinct from [Morris and Shin \(2002\)](#). There, the concern is that public information may be *too influential*; here, the planner may have *nothing useful to say* when saying something would matter most.

Trap-region value. Appendix [B](#), Proposition [8](#), shows that in the trap region in a Roman world, the expected value of truthful announcement under W^{tot} is $\mathcal{V}^{trap} = P^{esc} \cdot \Delta W^{esc} - P^{deepen} \cdot \Delta W^{deepen}$, with a precision threshold above which $\mathcal{V}^{trap} > 0$. Because ΔW^{esc} is large (escape avoids extended stagnation), even a moderately noisy signal can be worth announcing.

Non-monotone-region value. Proposition [9](#) establishes that in a Malthusian world, where the dynamics are non-monotone by Proposition [3](#), the value of announcement comes from reducing belief dispersion (and thereby dampening oscillations and accelerating convergence toward the interior steady state). This benefit is partially offset by adjustment costs from disrupting the existing dynamics.

The stabilizing role of Malthusians. An additional nuance distinguishes my setting from pure-complementarity coordination games. In response to a public announcement, Romanians coordinate (amplifying the signal), but Malthusians anti-coordinate (dampening the signal). This heterogeneity reduces the risk of over-coordination on noise. Compared to a population of only Romanians, a mixed population is safer to inform. The precision threshold for beneficial announcement is therefore *lower* in populations with substantial Malthusian minorities. Paradoxically, the presence of pessimists makes communication policy safer.

7.6 Comparison of Decentralized and Planner Allocations

Table [7](#) summarizes the contrast across criteria.

Table 7. Decentralized vs. Planner Allocations by Welfare Criterion

	Dynastic	Average Util.	Total Util.
<i>Romeroian World</i> ($\theta^* > 0$)			
Decentralized fertility		Too low (externality + wrong beliefs)	
Optimal announcement	Truthful	Mildly shaded toward θ_R	Strongly shaded toward θ_R
Selection-shifting role	None	Modest	Strongest
Residual inefficiency	Externality only	Reduced externality	Most reduced externality
<i>Malthusian World</i> ($\theta^* < 0$)			
Decentralized fertility		Too high (externality), mixed by beliefs	
Optimal announcement	Truthful	Mildly shaded toward θ_M	Strongly shaded toward θ_M
Selection-shifting role	None	Modest	Strongest
Residual inefficiency	Externality only	Reduced externality	Most reduced externality

Under all three criteria, the decentralized equilibrium is inefficient due to the uninternalized scale-effect externality. The dynastic planner can address belief distortions but not the externality. The average-utilitarian planner additionally uses the announcement as a partial substitute for an unavailable per-capita-fertility instrument. The total utilitarian planner shades further still, reflecting an additional motive to value population. Achieving full efficiency under any of the criteria would require additional instruments (fertility subsidies in Romerian worlds, fertility taxes in Malthusian worlds).

7.7 Policy Implications

The analysis yields several implications for government communication policy.

In the trap, take calculated risks. When the economy appears trapped—low fertility, stagnant population, persistent pessimism—even a moderately imprecise signal may be worth announcing if it has a chance of coordinating beliefs and initiating escape. Under the total utilitarian criterion, the expected value of escape justifies risk-taking.

Under dynastic welfare, be truthful. If the planner respects household preferences as given, the optimal policy is to provide accurate information. Distorting announcements would make some households worse off according to their own preferences, and the planner has nothing else to gain.

Under utilitarian criteria, shade systematically. The total utilitarian planner shades announcements toward optimism in Romerian worlds and toward pessimism in Malthusian worlds, by an amount that scales with the externality, the deviation of population from

the reference, and credibility. The average-utilitarian planner shades less but in the same direction.

Near the multiplicity boundary, the announcement does double duty. When the economy is approaching the multiplicity region (ϕ near ϕ^{**}), the announcement’s selection-shifting role becomes economically meaningful in addition to its information-provision role. The planner should account for both effects when calibrating the announcement.

During Malthusian non-monotone dynamics, time announcements carefully. In a Malthusian world, crises naturally correct beliefs through the learning channel of Proposition 3. Announcing during an expansion—when population is high and signals are informative—can accelerate learning and dampen the swings. Announcing during a crisis is less valuable because beliefs are already moving toward Malthusianism.

Build credibility for when it matters; recognize the limits of knowledge. The value of announcement depends on credibility. Planners should invest in credibility during normal times (by making accurate predictions) so that announcements are effective when intervention matters most. The most important policy implication, however, is epistemic humility. When the economy is in the trap, the planner knows the least about what to say. Acknowledging this uncertainty, rather than projecting false confidence, may be the most honest policy.

8 Conclusion

Beliefs about the economic consequences of population are not background assumptions in fertility models. They are first-order determinants of demographic and macroeconomic dynamics. This paper has formalized that observation in an overlapping generations model with heterogeneous beliefs about scale effects and a time cost of children, with multiplicative log-normal productivity dynamics that admit a tractable log-linear approximation around a balanced-growth reference. The resulting feedback between beliefs, fertility, and the informativeness of data produces a sharp duality. Malthusian minorities trap Romerian economies in stagnation, while Romerian minorities generate non-monotone dynamics in Malthusian economies. Wrong beliefs are always costly, but the form of the cost is determined by the truth they fail to track.

The strategic structure of the fertility game partitions the share space into three economically distinct regions: a Malthusian-dominated region of unconditional uniqueness, a

contraction region of conditional uniqueness covering most empirically relevant calibrations, and a Romerian-dominated multiplicity region whose existence depends on the strength of scale effects. A global-games refinement à la [Frankel et al. \(2003\)](#), applied to a reduction of the game in which the Malthusian best response has been substituted out, selects a risk-dominant equilibrium in the multiplicity region, restoring a unique selection across the entire share space. The trap region sits inside the contraction region for any standard calibration, so the trap intuition is robust to the multiplicity question. But where multiplicity is present, the planner’s announcement plays a selection-shifting role in addition to its information-provision role.

The informational poverty trap is the paper’s most distinctive theoretical object. Unlike traps grounded in technological nonconvexities or credit frictions, it arises endogenously from the geometry of Bayesian learning: signals are uninformative precisely in the region where pessimism keeps the economy confined, so the very beliefs that cause stagnation also insulate themselves from correction. This mechanism reframes a long-standing question in economic history. The pre-industrial Malthusian era can be read as a true Malthusian regime that ended with a technological break, or as a Romerian regime trapped for millennia by self-reinforcing pessimism that finally dissolved when population breached the informative region. The two readings carry sharply different implications for whether the modern fertility decline represents a benign demographic adjustment or a slow drift back into the trap.

Belief heterogeneity also forces a reckoning with how welfare itself is defined. Because the population over which welfare is evaluated is endogenous to policy, the dynastic, average-utilitarian, and total utilitarian criteria recommend systematically different communication strategies, with shading magnitudes that strictly increase as the planner’s normative commitments expand. The dynastic planner is fundamentally an information provider; the total utilitarian planner additionally values population. The same logic produces an unconventional message about public information: the planner is most uncertain exactly when announcements would be most valuable, and the heterogeneous strategic interaction softens but does not resolve this tension.

Several extensions are natural: a complete classification of the deterministic skeleton in Malthusian worlds; sharper causal identification of the belief–fertility link; integration with the quantity-quality margin central to unified growth theory; a microfoundation for announcement credibility; and a quantitative implementation that could in principle adjudicate between the two interpretations of pre-industrial history. Whether one reads human history as a Malthusian regime that ended or a Romerian regime that nearly never began, the lesson is the same: what societies believe about themselves shapes what they become, and what they become determines what they can learn.

References

- Angeletos, George-Marios and Jennifer La'O, "Sentiments," *Econometrica*, 2013, 81 (2), 739–779.
- , Fabrice Collard, and Harris Dellas, "Quantifying Confidence," *Econometrica*, 2018, 86 (5), 1689–1726.
- Ashraf, Quamrul and Oded Galor, "The "Out of Africa" Hypothesis, Human Genetic Diversity, and Comparative Economic Development," *American Economic Review*, 2013, 103 (1), 1–46.
- Azariadis, Costas and Allan Drazen, "Poverty Traps," *Quarterly Journal of Economics*, 1990, 105 (2), 501–526.
- Banerjee, Abhijit V and Andrew F Newman, "Occupational Choice and the Process of Development," *Journal of Political Economy*, 1993, 101 (2), 274–298.
- Barro, Robert J and Gary S Becker, "Fertility Choice in a Model of Economic Growth," *Econometrica*, 1989, 57 (2), 481–501.
- Becker, Gary S, "An Economic Analysis of Fertility," in "Demographic and Economic Change in Developed Countries," Columbia University Press, 1960, pp. 209–240.
- Bisin, Alberto and Thierry Verdier, "The Economics of Cultural Transmission and the Dynamics of Preferences," *Journal of Economic Theory*, 2001, 97 (2), 298–319.
- and —, "The Economics of Cultural Transmission and Socialization," in "Handbook of Social Economics," Vol. 1, Elsevier, 2011, pp. 339–416.
- Blackorby, Charles, Walter Bossert, and David Donaldson, *Population Issues in Social Choice Theory, Welfare Economics, and Ethics*, Cambridge University Press, 2005.
- Bloom, Nicholas, Charles I Jones, John Van Reenen, and Michael Webb, "Are Ideas Getting Harder to Find?," *American Economic Review*, 2020, 110 (4), 1104–1144.
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer, "Diagnostic Expectations and Credit Cycles," *Journal of Finance*, 2018, 73 (1), 199–227.
- Carlsson, Hans and Eric Van Damme, "Global Games and Equilibrium Selection," *Econometrica*, 1993, 61 (5), 989–1018.
- Cavalli-Sforza, Luigi Luca and Marcus W Feldman, *Cultural Transmission and Evolution: A Quantitative Approach*, Princeton University Press, 1981.
- Clark, Gregory, *A Farewell to Alms: A Brief Economic History of the World*, Princeton University Press, 2007.
- Daruich, Diego, "The Macroeconomic Consequences of Early Childhood Development Policies," Working Paper 2018-029, Federal Reserve Bank of St. Louis 2018.

- de la Croix, David and Matthias Doepke, “Inequality and Growth: Why Differential Fertility Matters,” *American Economic Review*, 2003, *93* (4), 1091–1113.
- Doepke, Matthias and Fabrizio Zilibotti, *Love, Money, and Parenting: How Economics Explains the Way We Raise Our Kids*, Princeton University Press, 2019.
- Frankel, David M, Stephen Morris, and Ady Pauzner, “Equilibrium selection in global games with strategic complementarities,” *Journal of Economic Theory*, 2003, *108* (1), 1–44.
- Galor, Oded, *Unified Growth Theory*, Princeton University Press, 2011.
- and David N Weil, “Population, Technology, and Growth: From Malthusian Stagnation to the Demographic Transition and Beyond,” *American Economic Review*, 2000, *90* (4), 806–828.
- Geanakoplos, John, “The Leverage Cycle,” *NBER Macroeconomics Annual*, 2010, *24* (1), 1–66.
- Goldstein, Itay and Ady Pauzner, “Demand-Deposit Contracts and the Probability of Bank Runs,” *Journal of Finance*, 2005, *60* (3), 1293–1327.
- Golosov, Mikhail, Larry E Jones, and Michèle Tertilt, “Efficiency with Endogenous Population Growth,” *Econometrica*, 2007, *75* (4), 1039–1071.
- Jones, Charles I, “R&D-Based Models of Economic Growth,” *Journal of Political Economy*, 1995, *103* (4), 759–784.
- , “The End of Economic Growth? Unintended Consequences of a Declining Population,” *American Economic Review*, 2022, *112* (11), 3489–3527.
- Kalemli-Ozcan, Sebnem, “A Stochastic Model of Mortality, Fertility, and Human Capital Investment,” *Journal of Development Economics*, 2003, *70* (1), 103–118.
- Kremer, Michael, “Population Growth and Technological Change: One Million BC to 1990,” *Quarterly Journal of Economics*, 1993, *108* (3), 681–716.
- Lee, Ronald, “The Demographic Transition: Three Centuries of Fundamental Change,” *Journal of Economic Perspectives*, 2003, *17* (4), 167–190.
- Lee, Sang Yoon (Tim) and Ananth Seshadri, “On the Intergenerational Transmission of Economic Status,” *Journal of Political Economy*, 2019, *127* (2), 855–921.
- Malmendier, Ulrike and Stefan Nagel, “Depression Babies: Do Macroeconomic Experiences Affect Risk Taking?,” *Quarterly Journal of Economics*, 2011, *126* (1), 373–416.
- Malthus, Thomas Robert, *An Essay on the Principle of Population*, J. Johnson, London, 1798.
- Manuelli, Rodolfo E and Ananth Seshadri, “Explaining International Fertility Differences,” *Quarterly Journal of Economics*, 2009, *124* (2), 771–807.

- Morris, Stephen and Hyun Song Shin, “Unique Equilibrium in a Model of Self-Fulfilling Currency Attacks,” *American Economic Review*, 1998, 88 (3), 587–597.
- and —, “Social Value of Public Information,” *American Economic Review*, 2002, 92 (5), 1521–1534.
- and —, “Global Games: Theory and Applications,” in Mathias Dewatripont, Lars Peter Hansen, and Stephen J Turnovsky, eds., *Advances in Economics and Econometrics: Theory and Applications, Eighth World Congress*, Vol. 1, Cambridge University Press, 2003, pp. 56–114.
- Parfit, Derek, *Reasons and Persons*, Oxford University Press, 1984.
- Romer, Paul M, “Endogenous Technological Change,” *Journal of Political Economy*, 1990, 98 (5, Part 2), S71–S102.
- Scheinkman, José A and Wei Xiong, “Overconfidence and Speculative Bubbles,” *Journal of Political Economy*, 2003, 111 (6), 1183–1220.
- Simsek, Alp, “Speculation and Risk Sharing with New Financial Assets,” *Quarterly Journal of Economics*, 2013, 128 (3), 1365–1396.
- Voigtländer, Nico and Hans-Joachim Voth, “The Three Horsemen of Riches: Plague, War, and Urbanization in Early Modern Europe,” *Review of Economic Studies*, 2013, 80 (2), 774–811.

A Proofs and Auxiliary Theoretical Results

A note on the appendix proofs

Several appendix results below rely on the local log-linear approximation (15) of the household value function. Throughout, I distinguish between (a) parts of the argument that are exact under the multiplicative log-normal technology of (3), and (b) parts that hold to first order around a balanced-growth reference. The proof of Proposition 3 is a direct calculation that does not rely on a dynamical-systems reduction; the global-games selection of Theorem 2 adapts the supermodular-game machinery of Frankel et al. (2003) to the reduced Romerian subgame obtained by substituting out the Malthusian best response.

A.1 Proof of Lemma 1

Step 1: Bellman equation. Substituting $c_{j,t} = A_t(1 - \psi(n))$ and using $\ln c_{j,t} = \ln A_t + \ln(1 - \psi(n))$ into the dynastic objective (6) yields the Bellman equation (14):

$$V_j(A, N, \phi, \pi) = \max_{n \geq 0} \left\{ \ln A + \ln(1 - \psi(n)) + \beta n \mathbb{E}_j[V_j(A', N', \phi', \pi')] \right\}.$$

The choice variable n enters the period return only through $\ln(1 - \psi(n))$ and the discount-multiplier βn ; aggregate state variables (A, N, ϕ, π) enter parametrically.

Step 2: Local log-linear conjecture. I conjecture the value function takes the local log-linear form

$$V_j(A, N, \phi, \pi) = v_j(\phi, \pi) + \xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N}) + R_j(A, N, \phi, \pi), \quad (31)$$

where R_j is a remainder of order $O(\|(\ln A - \ln A^*, \ln(N/\bar{N}))\|^2)$ around a balanced-growth reference $(A^*, \bar{N})$, and $v_j, \xi_{A,j}, \xi_{N,j}$ are determined by coefficient matching below. The remainder R_j is generated solely by the dependence of the optimal policy n_j^* on $(\ln A, \ln(N/\bar{N}))$ through the FOC; the technology side of the Bellman is exactly linear under (4), so it contributes no nonlinearity.

Step 3: Expected continuation under belief θ_j . Using (4),

$$\ln A' = \ln A + \bar{g} + \theta^* \ln(N/\bar{N}) + \eta_{t+1}.$$

Type- j households evaluate this expectation under their subjective belief θ_j (replacing θ^* by θ_j) and using $\mathbb{E}_j[\eta_{t+1}] = 0$:

$$\mathbb{E}_j[\ln A'] = \ln A + \bar{g} + \theta_j \ln(N/\bar{N}).$$

For population, with $N' = \bar{n}^e N$,

$$\mathbb{E}_j[\ln(N'/\bar{N})] = \ln(N/\bar{N}) + \ln \bar{n}^e.$$

Substituting into (31) and ignoring R_j at the reference,

$$\mathbb{E}_j[V'_j] = \mathbb{E}_j v_j(\phi', \pi') + \xi_{A,j} [\ln A + \bar{g} + \theta_j \ln(N/\bar{N})] + \xi_{N,j} [\ln(N/\bar{N}) + \ln \bar{n}^e] + O(\sigma_\eta^2), \quad (32)$$

where the $O(\sigma_\eta^2)$ term arises only if one wishes to retain the $\frac{1}{2}\xi_{A,j}^2\sigma_\eta^2$ Jensen correction. This Jensen term is a constant absorbed into v_j , so it does not affect the matching of $(\xi_{A,j}, \xi_{N,j})$.

Step 4: Coefficient matching at the reference. Substituting (32) into the Bellman (14), evaluating at the optimal policy n_j^* (treated as locally constant in $(\ln A, \ln(N/\bar{N}))$, with corrections absorbed in R_j):

$$v_j + \xi_{A,j} \ln A + \xi_{N,j} \ln(N/\bar{N}) = \ln A + \ln(1 - \psi(n_j^*)) + \beta n_j^* (\mathbb{E}_j v'_j + \xi_{A,j} [\ln A + \bar{g} + \theta_j \ln(N/\bar{N})] + \xi_{N,j} [\ln(N/\bar{N}) + \ln \bar{n}^e])$$

Matching coefficients on $\ln A$:

$$\xi_{A,j} = 1 + \beta n_j^* \xi_{A,j} \implies \xi_{A,j} = \frac{1}{1 - \beta n_j^*}.$$

Matching coefficients on $\ln(N/\bar{N})$:

$$\xi_{N,j} = \beta n_j^* (\xi_{A,j} \theta_j + \xi_{N,j}) \implies \xi_{N,j} (1 - \beta n_j^*) = \beta n_j^* \xi_{A,j} \theta_j = \frac{\beta n_j^* \theta_j}{1 - \beta n_j^*},$$

yielding

$$\xi_{N,j} = \frac{\beta n_j^* \theta_j}{(1 - \beta n_j^*)^2}.$$

This gives (17). The constant term v_j absorbs $\bar{g}\xi_{A,j}$, $\xi_{N,j} \ln \bar{n}^e$, the Jensen term, and the residual $\beta n_j^* \mathbb{E}_j v'_j$. The condition $\beta n_j^* < 1$ ensures $\xi_{A,j}$ and $\xi_{N,j}$ are finite.

Step 5: Fertility FOC. The first-order condition with respect to n in (14):

$$-\psi'(n) \cdot \frac{1}{1 - \psi(n)} + \beta \mathbb{E}_j[V'_j] = 0 \implies \frac{\psi'(n_j^*)}{1 - \psi(n_j^*)} = \beta \mathbb{E}_j[V'_j].$$

The left-hand side is strictly increasing in n by Assumption 2(iv), so the FOC has at most one solution. Under the specification (5), $\psi'(n) = \gamma n e^{-\gamma n^2/2}$ and $1 - \psi(n) = e^{-\gamma n^2/2}$, so $\psi'(n)/(1 - \psi(n)) = \gamma n$, and the FOC becomes

$$\gamma n_j^* = \beta \mathbb{E}_j[V'_j]. \quad (33)$$

Step 6: Existence and uniqueness of the fixed point. Substituting (32) into (33) and noting that $\xi_{A,j}$ and $\xi_{N,j}$ themselves depend on n_j^* , we obtain a fixed-point equation

$$\gamma n_j^* = \beta [\bar{v}_j(\phi, \pi) + \xi_{A,j}(n_j^*) (\ln A^* + \bar{g} + \theta_j \ln(N^*/\bar{N})) + \xi_{N,j}(n_j^*) (\ln(N^*/\bar{N}) + \ln \bar{n})]$$

at the reference $(A^*, N^*, \bar{n})$. Define the right-hand side as a function $F(n)$ of n (treating other variables as parameters). $F(n)$ is continuous on $[0, 1/\beta)$, with $F(0)$ finite and $F(n) \rightarrow \infty$ as $n \uparrow 1/\beta$ (because $\xi_{A,j}$ and $\xi_{N,j}$ blow up). The map $n \mapsto \gamma n - F(n)$ is therefore strictly increasing for n small enough and admits a unique root in $[0, \bar{n}^{\max}]$ for some $\bar{n}^{\max} < 1/\beta$ under the maintained calibration. Equilibrium fertility n_j^* is thus uniquely determined by θ_j and the reference state.

Step 7: Comparative statics in θ_j . Differentiating (33) totally with respect to θ_j at fixed $(\bar{n}, A^*, N^*, \phi, \pi)$:

$$\gamma \frac{\partial n_j^*}{\partial \theta_j} = \beta \left[\frac{\partial \xi_{A,j}}{\partial n_j^*} \frac{\partial n_j^*}{\partial \theta_j} \cdot [\dots] + \xi_{A,j} \ln(N^*/\bar{N}) + \frac{\partial \xi_{N,j}}{\partial \theta_j} + \frac{\partial \xi_{N,j}}{\partial n_j^*} \frac{\partial n_j^*}{\partial \theta_j} \cdot [\dots] \right].$$

At a balanced-growth reference where $\ln(N^*/\bar{N}) = 0$, the explicit- θ_j term reduces to $\beta \partial \xi_{N,j} / \partial \theta_j = \beta \cdot \beta n_j^* / (1 - \beta n_j^*)^2 > 0$. Provided the implicit terms (in n_j^*) do not flip the sign—which is verified by direct calculation under $\beta n_j^* < 1$ —this gives $\partial n_j^* / \partial \theta_j > 0$. Writing $n_j^* = n^* h(\theta_j; \bar{n})$ with h increasing in θ_j follows by reorganization. In particular, $\theta_R > \theta_M$ implies $n_R^* > n_M^*$. $\square$

A.2 Proof of Proposition 1

For type j expecting aggregate fertility $\bar{n}^e$, the conjecture (31) and equation (32) give

$$\mathbb{E}_j[V'_j] = v_j^{\text{const}} + \xi_{A,j} \ln A + \xi_{A,j} \theta_j \ln(N/\bar{N}) + \xi_{N,j} \ln(N/\bar{N}) + \xi_{N,j} \ln \bar{n}^e.$$

Hence

$$\frac{\partial \mathbb{E}_j[V'_j]}{\partial \bar{n}^e} = \frac{\xi_{N,j}}{\bar{n}^e}.$$

The FOC (33) reads $\gamma n_j^* = \beta \mathbb{E}_j[V'_j]$. Differentiating both sides totally with respect to $\bar{n}^e$, holding all other state variables fixed:

$$\gamma \frac{dn_j^*}{d\bar{n}^e} = \beta \cdot \frac{\xi_{N,j}}{\bar{n}^e},$$

where the dependence of $\xi_{N,j}$ on n_j^* generates a second-order correction that vanishes at the reference (because $\partial \xi_{N,j} / \partial n_j^*$ is multiplied by $dn_j^* / d\bar{n}^e$, giving an equation of the form $\gamma x = \beta \xi_{N,j} / \bar{n}^e + \beta (\partial \xi_{N,j} / \partial n) x \cdot [\dots]$, with the second term proportional to other state variables that vanish at the reference). Solving and substituting the closed form for $\xi_{N,j}$:

$$\frac{dn_j^*}{d\bar{n}^e} = \frac{\beta \xi_{N,j}}{\gamma \bar{n}^e} = \frac{\beta^2 n_j^* \theta_j}{\gamma \bar{n}^e (1 - \beta n_j^*)^2}.$$

This is (18). The sign matches the sign of θ_j : positive for Romerians and negative for Malthusians. $\square$

A.3 Proof of Theorem 1

Equilibrium condition. Let

$$BR(\bar{n}^e) = \phi n_R^*(\bar{n}^e) + (1 - \phi) n_M^*(\bar{n}^e).$$

A rational-expectations equilibrium is a fixed point $\bar{n}^* = BR(\bar{n}^*)$. The local slope of BR at any candidate fixed point is

$$\rho(\phi) = BR'(\bar{n}^*) = \phi d_R + (1 - \phi) d_M,$$

with $d_R > 0$ and $d_M < 0$ by Proposition 1.

Part (i): Malthusian-dominated region ($\phi < \phi^*$). For $\phi < \phi^* = |d_M| / (d_R + |d_M|)$, the slope satisfies $\rho(\phi) < 0$ by direct computation. BR is therefore locally strictly decreasing in

a neighborhood of any candidate fixed point. Define the residual map $\Phi(x) = BR(x) - x$, which is continuously differentiable with $\Phi'(x) = \rho(\phi) - 1 < -1 < 0$ at any fixed point. By the implicit function theorem, fixed points of BR are isolated. Moreover, since both $n_R^*(\bar{n}^e)$ and $n_M^*(\bar{n}^e)$ are continuous and bounded above by $\bar{n}^{\max} < 1/\beta$, BR maps $[0, \bar{n}^{\max}]$ into itself, so by the intermediate value theorem applied to Φ , at least one fixed point exists. Strict monotonicity (decreasing) of Φ (since $\Phi' < 0$ everywhere on the candidate region) implies the fixed point is unique on $[0, \bar{n}^{\max}]$. The argument requires only the sign of $\rho(\phi)$, not its magnitude; uniqueness is therefore unconditional in this region.

Part (ii): Contraction region ($\phi^* \leq \phi \leq \phi^{**}$, or all $\phi \geq \phi^*$ if $d_R \leq 1$). For ϕ in this range, $0 \leq \rho(\phi) < 1$. The map $BR : [0, \bar{n}^{\max}] \rightarrow [0, \bar{n}^{\max}]$ is therefore Lipschitz with constant $\rho(\phi) < 1$ on a neighborhood of any candidate fixed point. By Banach's fixed-point theorem applied to a closed neighborhood, the fixed point is unique. When $d_R \leq 1$, $\phi^{**} = (1 + |d_M|)/(d_R + |d_M|) \geq 1$, so the contraction region extends to all $\phi \in [\phi^*, 1]$ and there is no multiplicity region.

Part (iii): Multiplicity region ($\phi > \phi^{**}$; requires $d_R > 1$). For $\phi > \phi^{**}$, $\rho(\phi) > 1$. Locally near any fixed point, $\Phi'(x) = \rho - 1 > 0$, so BR crosses the 45-degree line from below with positive slope greater than one. Since BR is continuous on $[0, \bar{n}^{\max}]$ with $BR(0) \geq 0$ and $BR(\bar{n}^{\max}) \leq \bar{n}^{\max}$ (otherwise fertility would diverge), Φ takes nonnegative values near 0 and nonpositive values near $\bar{n}^{\max}$. By the intermediate value theorem combined with the local slope condition, the number of transversal intersections is odd and at least three (if any). Standard transversality arguments imply the local slopes alternate between greater than one (unstable) and less than one (stable) as one moves through consecutive fixed points. The extreme intersections are stable; an odd number ≥ 3 of equilibria therefore typically arises with the extreme two stable.

Multiplicity condition. $\phi^{**} < 1$ iff $(1 + |d_M|) < (d_R + |d_M|)$ iff $d_R > 1$. Substituting (18) for d_R gives

$$\frac{\beta^2 n_R^* \theta_R}{\gamma \bar{n} (1 - \beta n_R^*)^2} > 1 \iff \theta_R > \theta_R^{crit} \equiv \frac{\gamma \bar{n} (1 - \beta n_R^*)^2}{\beta^2 n_R^*}. \quad \square$$

A.4 Proof of Theorem 2

I adapt the supermodular global-games argument of Frankel et al. (2003) via a reduction that substitutes out the Malthusian best response.

Step 1: Reduction to a Romerian subgame. Let n_R^i denote individual Romerian fertility choices and let $\bar{n}_R = \int_R n_R^i di / \phi$ denote average Romerian fertility. The aggregate fertility identity is

$$\bar{n} = \phi \bar{n}_R + (1 - \phi) n_M^*(\bar{n}).$$

Because Malthusians have a strictly downward-sloping best response with slope $d_M < 0$, the map $\bar{n} \mapsto \phi \bar{n}_R + (1 - \phi) n_M^*(\bar{n})$ for fixed $\bar{n}_R$ has derivative $(1 - \phi) d_M < 0$, and so the equation $\bar{n} = \phi \bar{n}_R + (1 - \phi) n_M^*(\bar{n})$ has a unique solution $\bar{n} = T(\bar{n}_R)$ for each $\bar{n}_R$. By the implicit function theorem,

$$T'(\bar{n}_R) = \frac{\phi}{1 - (1 - \phi) d_M} > 0.$$

The reduced game has Romerian players choosing n_R^i , with payoffs $\tilde{U}_R(n_R^i, \bar{n}_R) := U_R(n_R^i, T(\bar{n}_R))$. The Malthusian players are eliminated as strategic actors; their best response is built into T .

Step 2: Supermodularity of the reduced game. The cross-partial of the reduced payoff is

$$\frac{\partial^2 \tilde{U}_R}{\partial n_R^i \partial \bar{n}_R} = \frac{\partial^2 U_R}{\partial n_R^i \partial \bar{n}} \cdot T'(\bar{n}_R).$$

By Proposition 1, $\partial^2 U_R / \partial n_R^i \partial \bar{n} > 0$, and $T'(\bar{n}_R) > 0$ by Step 1. Therefore $\partial^2 \tilde{U}_R / \partial n_R^i \partial \bar{n}_R > 0$: the reduced game is strictly supermodular in actions among Romerians.

Step 3: Monotone signal structure and dominance regions. Each Romerian observes $\phi_i = \phi + \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, \sigma_\zeta^2)$ i.i.d. With a diffuse improper prior on ϕ (or any sufficiently flat proper prior), the conditional distribution of ϕ given ϕ_i is also Gaussian with mean ϕ_i and variance σ_ζ^2 . Higher ϕ_i corresponds to first-order stochastically higher posteriors over ϕ , satisfying the monotone likelihood ratio property. Under the calibration of the multiplicity region, there exist values $\bar{\phi}_L < \phi^{**}$ and $\bar{\phi}_H > 1$ such that for $\phi < \bar{\phi}_L$ the low-fertility selection strictly dominates regardless of others' actions, and for $\phi > \bar{\phi}_H$ the high-fertility selection strictly dominates. (Since $\phi \in [0, 1]$, the upper dominance region is implemented by extending the action space or by treating $\bar{\phi}_H$ as the conditioning event under which the unique global-games equilibrium plays the high-fertility selection.)

Step 4: Application of Frankel et al. (2003). The reduced Romerian game now satisfies the maintained conditions of Theorem 1 of Frankel et al. (2003): (a) supermodular payoffs in actions; (b) state-monotone payoffs (higher ϕ shifts the cross-partial upward, since T' is increasing in ϕ by direct computation); (c) continuous payoffs; (d) dominance regions

on both ends of the state space. Theorem 1 of Frankel–Morris–Pauzner therefore applies: as $\sigma_\zeta \rightarrow 0$, iterated deletion of strictly dominated strategies converges to a unique strategy profile in monotone (threshold) form. Each Romerian uses a cutoff rule $\phi_i \gtrless \phi^{rd} \Rightarrow \text{high-}/\text{low-fertility selection}$.

Step 5: Closed-form risk-dominant cutoff. At the cutoff $\phi_i = \phi^{rd}$, a Romerian must be indifferent between the two selections under Laplacian beliefs over the share of others choosing high (which is uniform on $[0, 1]$ in the limit $\sigma_\zeta \rightarrow 0$; see [Morris and Shin \(2003\)](#)). Let V_R^L and V_R^H denote the continuation values under the low- and high-fertility selections, respectively, with corresponding aggregate fertilities $\bar{n}^L$ and $\bar{n}^H$. The indifference condition reduces to

$$\int_0^1 [U_R(n_R^*(\bar{n}_q), \bar{n}_q) - U_R^0] dq = 0,$$

where $\bar{n}_q$ is the aggregate fertility implied by a fraction q of Romerians playing high, and U_R^0 is a normalization. Equivalently, using monotonicity in $\bar{n}^e$ and the FOC (33), the cutoff solves

$$\int_{\bar{n}^L}^{\bar{n}^H} u'_R(n_R^*(\bar{n}^e)) d\bar{n}^e = 2 \int_{\bar{n}^L}^{\bar{n}^{rd}} u'_R(n_R^*(\bar{n}^e)) d\bar{n}^e,$$

which under the log-quadratic specification has a unique closed-form solution $\bar{n}^{rd}$, and ϕ^{rd} is recovered from $T^{-1}(\bar{n}^{rd}) = \phi \bar{n}_R^{rd} + (1 - \phi)n_M^*(\bar{n}^{rd})$. The selected equilibrium is risk-dominant in the sense of [Frankel et al. \(2003\)](#). $\square$

A.5 Proof of Theorem 3

Step 1: Demographic-selection drift. For interior ϕ_t ,

$$\phi_t^{birth} - \phi_t = \frac{\phi_t n_{R,t}}{\bar{n}_t} - \phi_t = \frac{\phi_t(n_{R,t} - \bar{n}_t)}{\bar{n}_t} = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t} > 0$$

(suppressing the time- t star superscripts on n_j^* for readability). This drift is *strictly* positive at every interior ϕ_t and has no information dependence.

Step 2: Posterior dynamics under truth. Let $L_t = \ln[(1 - \pi_t)/\pi_t]$ be the log-odds in favor of Malthus. Bayes' rule (11) gives

$$L_{t+1} - L_t = \ln \frac{\ell(s_t | \theta_M, N_t)}{\ell(s_t | \theta_R, N_t)} = \frac{(\theta_R - \theta_M) \ln(N_t/\bar{N})}{\sigma_\eta^2} \left[s_t - \bar{g} - \frac{\theta_R + \theta_M}{2} \ln(N_t/\bar{N}) \right],$$

which follows from expanding the log-likelihood ratio of two Gaussians with different means but the same variance. Under the truth $\theta^* = \theta_R$, $\mathbb{E}^{\theta_R}[s_t | \mathcal{F}_t] = \bar{g} + \theta_R \ln(N_t/\bar{N})$, so

$$\mathbb{E}^{\theta_R}[L_{t+1} - L_t | \mathcal{F}_t] = \frac{(\theta_R - \theta_M) \ln(N_t/\bar{N})}{\sigma_\eta^2} \cdot \frac{(\theta_R - \theta_M) \ln(N_t/\bar{N})}{2} \cdot (-1) \cdot 1 = -\mathcal{I}_t \leq 0.$$

Therefore L_t is a supermartingale under truth (in expectation, decreasing). By Doob's martingale convergence theorem, L_t converges a.s. to some $[-\infty, \infty]$ -valued limit L_∞ .

Step 3: $\sum_t \mathcal{I}_t = \infty$ almost surely. I show this by contradiction. Suppose $\sum_t \mathcal{I}_t < \infty$ on a set of positive probability. Then $\mathcal{I}_t \rightarrow 0$, implying $\ln(N_t/\bar{N}) \rightarrow 0$. On this event, eventually N_t is in any neighborhood of $\bar{N}$; equivalently, $\bar{n}_t \rightarrow 1$ along a subsequence of full asymptotic density.

By Assumption 4, $\bar{n}_t \rightarrow 1$ requires $\phi_t \rightarrow \phi^{trap}$. But on the same event, the cumulative posterior increments are summable, so $L_t \rightarrow L_\infty$ finite, and through (13), the cumulative cultural-transmission drift on ϕ_t accumulates without correction:

$$\phi_{t+1} - \phi_t \geq \lambda(\phi_t^{birth} - \phi_t) - (1 - \lambda) |\Delta\pi_{t+1}|,$$

with $\sum |\Delta\pi_{t+1}| < \infty$. Selection drift $\lambda(\phi_t^{birth} - \phi_t)$ is bounded below by λc for some $c > 0$ whenever $\phi_t \in [\phi^{trap} - \varepsilon, \phi^{trap} + \varepsilon]$ and $\phi_t < 1$ (this holds because $\phi_t(1 - \phi_t)(n_R - n_M)/\bar{n}_t$ is uniformly positive on a compact interior subset). Summing, ϕ_t increases beyond any bound below 1, contradicting $\phi_t \rightarrow \phi^{trap} < 1$. Hence $\sum_t \mathcal{I}_t = \infty$ a.s.

Step 4: $\pi_t \rightarrow 1$ almost surely. Combining $\mathbb{E}^{\theta_R}[\Delta L_t] = -\mathcal{I}_t$ with $\sum \mathcal{I}_t = \infty$, the law of large numbers for martingale differences (Azuma–Hoeffding-style or its Lévy variant) gives $L_t \rightarrow -\infty$ a.s., hence $\pi_t \rightarrow 1$ a.s.

Step 5: $\phi_t \rightarrow 1$ almost surely. Given $\pi_t \rightarrow 1$, for any $\varepsilon > 0$ there exists $T(\omega)$ with $\pi_t > 1 - \varepsilon$ for $t \geq T$. The dynamics (13) give

$$1 - \phi_{t+1} = \lambda(1 - \phi_t^{birth}) + (1 - \lambda)(1 - \pi_{t+1}) \leq \lambda(1 - \phi_t) + (1 - \lambda)\varepsilon,$$

where the inequality uses $\phi_t^{birth} \geq \phi_t$. Iterating,

$$\limsup_{t \rightarrow \infty} (1 - \phi_t) \leq \frac{(1 - \lambda)\varepsilon}{1 - \lambda} = \varepsilon.$$

Since ε is arbitrary, $\phi_t \rightarrow 1$ a.s.

Step 6: $\bar{n}_t \rightarrow n_R^*$ and $\sup_t \ln(N_t/\bar{N}) = +\infty$. Convergence of fertility follows from continuity of n_j^* in the state and $\phi_t \rightarrow 1$. Since $n_R^* > 1$ by Assumption 4, eventually $\bar{n}_t \geq (1 + n_R^*)/2 > 1$, so N_t grows geometrically, and $\ln(N_t/\bar{N}) \rightarrow +\infty$ a.s. $\square$

A.6 Proof of Proposition 2

Throughout, the trajectory is assumed to remain in $\mathcal{T}(\delta, \zeta)$; the proof also verifies trap closure.

Part (i): Bound on $\mathbb{E}[\Delta\pi_t \mid \mathcal{F}_t]$. Using the chain rule on $\pi_t = (1 + e^{L_t})^{-1}$,

$$\frac{d\pi}{dL} = -\pi(1 - \pi).$$

A second-order Taylor expansion in ΔL :

$$\Delta\pi_t = -\pi_t(1 - \pi_t) \Delta L_t + \frac{1}{2}\pi_t(1 - \pi_t)(1 - 2\pi_t)(\Delta L_t)^2 + O((\Delta L_t)^3).$$

Taking expectations under truth (here $\theta^* = \theta_R$) and using $\mathbb{E}[(\Delta L_t)^2] = O(\mathcal{I}_t)$:

$$|\mathbb{E}[\Delta\pi_t \mid \mathcal{F}_t]| \leq \pi_t(1 - \pi_t) \mathcal{I}_t + O(\mathcal{I}_t).$$

In the trap region, $\mathcal{I}_t \leq \bar{\mathcal{I}}(\delta) = (\theta_R - \theta_M)^2 \delta^2 / (2\sigma_\eta^2)$. Since $\pi(1 - \pi) \leq 1/4$, the leading term is bounded by $\bar{\mathcal{I}}(\delta)/4 = C_1 \bar{\mathcal{I}}(\delta)$ with $C_1 = 1/4$, absorbing the higher-order term into the constant for δ small.

Part (ii): Bound on demographic-selection drift. By Step 1 of the previous proof,

$$\phi_{t+1}^{birth} - \phi_t = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t} \leq \frac{(1 - \phi_t)(n_R - n_M)}{n_M},$$

where the inequality uses $\phi_t/\bar{n}_t \leq 1/n_M$ (since $\bar{n}_t = \phi_t n_R + (1 - \phi_t)n_M \geq \phi_t n_M + (1 - \phi_t)n_M = n_M$ requires $n_R \geq n_M$, but stronger: $\bar{n}_t \geq \phi_t n_M$ needs $n_R \geq n_M$, which holds). Therefore $C_2 = 1/n_M$.

Trap closure. I verify that if $(\phi_t, N_t, \pi_t) \in \mathcal{T}(\delta, \zeta)$, then $(\phi_{t+1}, N_{t+1}, \pi_{t+1}) \in \mathcal{T}(\delta, \zeta)$ for δ, ζ small enough (with positive probability close to 1; the strict statement is in expectation). The condition $\bar{n}(\phi_{t+1}) \in [1, 1 + \zeta]$ follows from continuity and the bounded drift in ϕ_t . The condition $|\ln(N_{t+1}/\bar{N})| < \delta$ follows from $\bar{n}_t \in [1, 1 + \zeta]$, so N_t grows by at most $(1 + \zeta)$ per period and stays in the δ -neighborhood for ζ sufficiently small. The condition $\pi_{t+1} \leq \phi_{t+1}$:

note

$$\phi_{t+1} - \pi_{t+1} = \lambda(\phi_t^{birth} - \pi_{t+1}) = \lambda(\phi_t^{birth} - \pi_t) - \lambda(\pi_{t+1} - \pi_t) = \lambda(\phi_t^{birth} - \phi_t) + \lambda(\phi_t - \pi_t) - \lambda \Delta \pi_t.$$

Each summand is non-negative in expectation under truth in the trap (selection ≥ 0 , $\phi_t \geq \pi_t$, and $\Delta \pi_t \geq 0$ on average since truth is Romerian). Hence $\phi_{t+1} \geq \pi_{t+1}$ in expectation; the strict statement follows by induction with high probability.

Part (iii): Lower bound on $\mathbb{E}[T^{esc}]$. Combining (i) and (ii), the joint drift in ϕ_t within the trap satisfies

$$\mathbb{E}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] \leq \lambda C_2(n_R - n_M)(1 - \phi_t) + (1 - \lambda)C_1\bar{\mathcal{I}}(\delta) \leq D,$$

where $D = \lambda C_2(n_R - n_M) + (1 - \lambda)C_1\bar{\mathcal{I}}(\delta)$ (using $1 - \phi_t \leq 1$).

Define $M_t = \phi_t - tD$. Then

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = \mathbb{E}[\phi_{t+1} \mid \mathcal{F}_t] - (t+1)D \leq \phi_t + D - (t+1)D = M_t.$$

M_t is therefore a supermartingale on the trap event. Let $T = T^{esc}$. For each finite n , optional stopping applied to the bounded stopping time $T \wedge n$:

$$\mathbb{E}[M_{T \wedge n}] \leq M_0 = \phi_0.$$

On $\{T \leq n\}$, $\phi_T > \bar{\phi}$ by definition; on $\{T > n\}$, $\phi_n \geq 0$. Hence

$$\mathbb{E}[\phi_{T \wedge n}] - D \mathbb{E}[T \wedge n] \leq \phi_0,$$

$$D \mathbb{E}[T \wedge n] \geq \mathbb{E}[\phi_{T \wedge n}] - \phi_0 \geq \bar{\phi} \mathbb{P}(T \leq n) - \phi_0.$$

Letting $n \rightarrow \infty$ by monotone convergence, and assuming $\mathbb{P}(T < \infty) = 1$ (which holds by Theorem 3),

$$D \mathbb{E}[T] \geq \bar{\phi} - \phi_0, \quad \text{i.e.,} \quad \mathbb{E}[T^{esc}] \geq \frac{\bar{\phi} - \phi_0}{D}. \quad \square$$

A.7 Trap Duration (Auxiliary Result)

Corollary 3. *Under Proposition 2, when learning is the binding escape channel, $\mathbb{E}[T^{esc}] = \Omega(1/\delta^2)$ as $\delta \rightarrow 0$.*

Proof. $\bar{\mathcal{I}}(\delta) = (\theta_R - \theta_M)^2 \delta^2 / (2\sigma_\eta^2) = O(\delta^2)$. With $D = \lambda C_2(n_R - n_M) + (1 - \lambda)C_1\bar{\mathcal{I}}(\delta) = a + b\delta^2$ for constants $a = \lambda C_2(n_R - n_M)$ and $b = (1 - \lambda)C_1(\theta_R - \theta_M)^2 / (2\sigma_\eta^2)$, we have

$\mathbb{E}[T^{esc}] \geq (\bar{\phi} - \phi_0)/(a + b\delta^2)$. When the cultural channel is shut down ($\lambda = 0$) or weak relative to the learning channel, a is small and $\mathbb{E}[T^{esc}] \geq \Omega(\delta^{-2})$. $\square$

A.8 Trap Boundary (Auxiliary Result)

Proposition 7. *The critical population at which the learning channel matches the demographic-selection channel in magnitude is*

$$\frac{N^{crit}(\phi)}{\bar{N}} = \exp \left(\sqrt{\frac{2\sigma_\eta^2 \cdot \lambda C_2(n_R - n_M)(1 - \phi)}{(1 - \lambda)C_1(\theta_R - \theta_M)^2}} \right).$$

Proof. Equate the per-period selection drift bound $\lambda C_2(n_R - n_M)(1 - \phi)$ with the per-period learning drift $(1 - \lambda)C_1(\theta_R - \theta_M)^2[\ln(N/\bar{N})]^2/(2\sigma_\eta^2)$ and solve for $\ln(N/\bar{N})$. $\square$

A.9 Proof of Proposition 3

Part (i): Selection drift on births. As in Theorem 3,

$$\phi_t^{birth} - \phi_t = \frac{\phi_t(1 - \phi_t)(n_R - n_M)}{\bar{n}_t}.$$

Since $\phi_t \in (0, 1)$ implies $\phi_t(1 - \phi_t) \in (0, 1/4]$, and $\bar{n}_t \geq n_M > 0$, the drift is strictly positive and bounded above by $(n_R - n_M)/(4n_M)$. It does not depend on $\ln(N_t/\bar{N})$.

Part (ii): Learning drift on the posterior. Repeating the log-likelihood-ratio expansion of Theorem 3 but under the truth $\theta^* = \theta_M$:

$$L_{t+1} - L_t = \frac{(\theta_R - \theta_M) \ln(N_t/\bar{N})}{\sigma_\eta^2} \left[s_t - \bar{g} - \frac{\theta_R + \theta_M}{2} \ln(N_t/\bar{N}) \right].$$

Under $\theta^* = \theta_M$, $\mathbb{E}^{\theta_M}[s_t] = \bar{g} + \theta_M \ln(N_t/\bar{N})$. Substituting,

$$\mathbb{E}^{\theta_M}[L_{t+1} - L_t \mid \mathcal{F}_t] = \frac{(\theta_R - \theta_M) \ln(N_t/\bar{N})}{\sigma_\eta^2} \cdot [\theta_M \ln(N_t/\bar{N}) - \frac{\theta_R + \theta_M}{2} \ln(N_t/\bar{N})] = +\mathcal{I}_t.$$

So $\mathbb{E}^{\theta_M}[\Delta L_t] = +\mathcal{I}_t \geq 0$. Using the chain-rule expansion $d\pi/dL = -\pi(1 - \pi)$,

$$\mathbb{E}^{\theta_M}[\pi_{t+1} - \pi_t \mid \mathcal{F}_t] = -\pi_t(1 - \pi_t) \mathcal{I}_t + o(\mathcal{I}_t).$$

This is *strictly negative* for $\pi_t \in (0, 1)$ and $\ln(N_t/\bar{N}) \neq 0$, and exactly zero at $\ln(N_t/\bar{N}) = 0$ (since $\mathcal{I}_t = 0$ there). Quadratic dependence on population deviation is immediate from the

form of $\mathcal{I}_t$.

Part (iii): Sign change of the joint drift in ϕ_t . From the law of motion $\phi_{t+1} = \lambda\phi_t^{birth} + (1-\lambda)\pi_{t+1}$,

$$\phi_{t+1} - \phi_t = \lambda(\phi_t^{birth} - \phi_t) + (1-\lambda)(\pi_{t+1} - \phi_t).$$

Adding and subtracting $(1-\lambda)\pi_t$,

$$\phi_{t+1} - \phi_t = \lambda(\phi_t^{birth} - \phi_t) + (1-\lambda)(\pi_t - \phi_t) + (1-\lambda)(\pi_{t+1} - \pi_t).$$

Taking expectations under truth θ_M and using Part (ii):

$$\mathbb{E}^{\theta_M}[\phi_{t+1} - \phi_t \mid \mathcal{F}_t] = \underbrace{\lambda(\phi_t^{birth} - \phi_t)}_{>0} + \underbrace{(1-\lambda)(\pi_t - \phi_t)}_{\text{sign of } \pi_t - \phi_t} + \underbrace{(1-\lambda)[- \pi_t(1-\pi_t)\mathcal{I}_t + o(\mathcal{I}_t)]}_{\leq 0}. \quad (34)$$

At $\ln(N_t/\bar{N}) = 0$: $\mathcal{I}_t = 0$, so the third term vanishes. The drift reduces to $\lambda(\phi_t^{birth} - \phi_t) + (1-\lambda)(\pi_t - \phi_t)$. This is non-negative whenever

$$\pi_t \geq \phi_t - \frac{\lambda}{1-\lambda}(\phi_t^{birth} - \phi_t),$$

which is satisfied in particular when $\pi_t \geq \phi_t$. When λ is close to 1 (strong cultural transmission), the inequality holds for any π_t close enough to ϕ_t .

For $|\ln(N_t/\bar{N})|$ large: the third term grows quadratically in $\ln(N_t/\bar{N})$, while the first two terms are bounded independently of $\ln(N_t/\bar{N})$. The third term therefore dominates and the drift becomes strictly negative once

$$\pi_t(1-\pi_t)\mathcal{I}_t > \frac{\lambda(\phi_t^{birth} - \phi_t)}{1-\lambda} + \max\{\pi_t - \phi_t, 0\}.$$

Substituting $\mathcal{I}_t = (\theta_R - \theta_M)^2[\ln(N_t/\bar{N})]^2/(2\sigma_\eta^2)$ and solving for $|\ln(N_t/\bar{N})|$:

$$[\ln(N_t/\bar{N})]^2 > \bar{L}(\phi_t, \pi_t, \lambda)^2 \equiv \frac{2\sigma_\eta^2[\lambda(\phi_t^{birth} - \phi_t)/(1-\lambda) + \max\{\pi_t - \phi_t, 0\}]}{\pi_t(1-\pi_t)(\theta_R - \theta_M)^2}.$$

Continuity of the drift in $\ln(N_t/\bar{N})$ implies the existence of a sign-changing state. $\square$

A.10 State-Dependent Value of Announcement (Auxiliary Results)

Proposition 8. *Suppose the economy is in $\mathcal{T}(\delta, \zeta)$ in a Romerian world. Under W^{tot} ,*

$$\mathcal{V}^{trap}(\tau_\nu^{trap}) = P^{esc}(\tau_\nu^{trap}) \cdot \Delta W^{esc} - P^{deepen}(\tau_\nu^{trap}) \cdot \Delta W^{deepen},$$

with a threshold $\underline{\tau}^{trap}$ above which $\mathcal{V}^{trap} > 0$. Because ΔW^{esc} is large, $\underline{\tau}^{trap}$ is moderate.

Sketch of proof. An announcement of precision τ_ν shifts households' subjective θ^{post} toward the announced value, with magnitude proportional to $\tau_\nu/(\tau_\nu + \tau_{prior})$. In the trap, the Romerian share ϕ_t is below ϕ^{trap} . A truthful announcement raises subjective optimism, increasing n_R^* and n_M^* (the latter increase smaller because Malthusians are pessimists). If the resulting $\bar{n}_t$ exceeds 1, population grows, learning engages, and escape probability P^{esc} becomes positive. The deepening event P^{deepen} corresponds to the announcement-induced shift being insufficient to escape but sufficient to move the trajectory deeper into the trap; under truthful announcements in a Romerian world this event has probability bounded above by $\mathbb{P}(\nu < -\nu_{\min})$ for some threshold $\nu_{\min} > 0$, which is small for moderate τ_ν . The threshold $\underline{\tau}^{trap}$ is the value of τ_ν at which $P^{esc} \Delta W^{esc} = P^{deepen} \Delta W^{deepen}$; since ΔW^{esc} reflects the large welfare gain of escaping prolonged stagnation while ΔW^{deepen} is bounded by the marginal change in stagnation duration, $\underline{\tau}^{trap}$ is moderate. $\square$

Proposition 9. *In a Malthusian world with non-monotone dynamics under W^{tot} ,*

$$\mathcal{V}^{cycle}(\tau_\nu) = \Omega \left(\text{Var}(\theta_i) - \mathbb{E}[\text{Var}(\theta_i^{post})] \right) - \Lambda \left((\bar{\theta} - \theta^*)^2 - \mathbb{E}[(\bar{\theta}^{post} - \theta^*)^2] \right) - C^{adj},$$

where $\Omega > 0$ is the marginal welfare gain from reducing dispersion, $\Lambda > 0$ is the marginal welfare gain from reducing mean bias, and $C^{adj} > 0$ is an adjustment cost. For sufficiently precise signals, $\mathcal{V}^{cycle} > 0$.

Sketch of proof. The decomposition (28) provides the mapping from variance and bias of beliefs to welfare losses Δ_3 and Δ_2 . The expectations are taken under the Bayesian posterior obtained from the announcement; both the variance and the bias are decreasing in τ_ν . The adjustment cost reflects the welfare loss from disrupting agents' contingent plans during a cycle. $\square$

B Welfare Analysis Proofs and Auxiliary Results

B.1 Proof of Proposition 4

Step 1: Planner's Bellman. Define the planner value function for the total utilitarian objective (26):

$$V^P(A_t, N_t) = \max_{n_t} \left\{ N_t [\ln A_t + \ln(1 - \psi(n_t))] + \beta \mathbb{E}^*[V^P(A_{t+1}, n_t N_t)] \right\}.$$

Conjecture $V^P(A, N) = N \cdot w^P(A, N)$ with $w^P(A, N) \approx v^P + \xi_A^P \ln A + \xi_N^P \ln(N/\bar{N})$ at a balanced-growth reference. Substituting,

$$N w^P(A, N) = N [\ln A + \ln(1 - \psi(n))] + \beta \mathbb{E}^*[n N w^P(A', nN)].$$

Dividing by N ,

$$w^P(A, N) = \ln A + \ln(1 - \psi(n)) + \beta n \mathbb{E}^*[w^P(A', N')].$$

Note the planner's per-capita Bellman has the same form as the household's, but the planner's FOC will additionally internalize the population-multiplier through differentiating $N_{t+1} = nN_t$.

Step 2: Coefficient matching. Repeating the matching of Lemma 1 but with θ^* (truth) instead of θ_j :

$$\xi_A^P = \frac{1}{1 - \beta n^{FB}}, \quad \xi_N^P = \frac{\beta n^{FB} \theta^*}{(1 - \beta n^{FB})^2}.$$

Step 3: Planner FOC. Differentiating the planner's total welfare objective with respect to n_t and noting $N_{t+1} = n_t N_t$ so $\partial V^P / \partial N_{t+1} = w^P + N_{t+1} \partial w^P / \partial N$, at the reference $\partial w^P / \partial N = \xi_N^P / N$, so the marginal value of population is $V_N^P = w^P + \xi_N^P$. The FOC reads

$$\frac{\psi'(n_t^{FB})}{1 - \psi(n_t^{FB})} = \beta [w^P(A', N') + \xi_N^P],$$

which is (27). Under $\psi(n) = 1 - e^{-\gamma n^2/2}$, this is $\gamma n^{FB} = \beta [w^P + \xi_N^P]$.

Step 4: Comparison with household. The household's common-correct-belief FOC is $\gamma n^{CB} = \beta \mathbb{E}[V_{CB}(A', N')]$, where $V_{CB}(A, N) = w^P(A, N)$ (same per-capita value function under correct beliefs because the household value function with $\theta = \theta^*$ coincides with the

planner's per-capita value). The wedge between (27) and the household FOC is exactly $\beta\xi_N^P$, with the sign of θ^* . Strict monotonicity of the LHS γn gives:

$$\theta^* > 0 \implies \xi_N^P > 0 \implies n^{FB} > n^{CB}, \quad \theta^* < 0 \implies n^{FB} < n^{CB}.$$

Step 5: $n^{CB} > n_M^*$. By Lemma 1, fertility is increasing in θ_j . With $\theta^* = \theta_R > 0$, $n^{CB} = n^*(\theta_R) > n_M^* = n^*(\theta_M)$. Combined with Step 4, $n^{FB} > n^{CB} > n_M^*$ in a Romerian world.

Step 6: Welfare decomposition. The decomposition (28) of $W^{FB} - W^{HB}$ into externality, mean-bias, and dispersion components is established below. $\square$

B.2 First-Best under Average Utilitarianism (Auxiliary Result)

Proposition 10. *In a Romerian world, $n^{CB} < n^{FB,avg} < n^{FB,tot}$. The average-utilitarian planner internalizes the per-capita scale-effect channel through $\ln A_t$, but lacks the population-multiplier term N_t that gives the total utilitarian planner the additional wedge.*

Proof. The average-utilitarian Bellman is $V^{P,avg}(A, N) = \ln A + \ln(1 - \psi(n)) + \beta\mathbb{E}^*[V^{P,avg}(A', N')]$, without the multiplicative N . The matching gives $\xi_A^{P,avg} = 1/(1 - \beta n^{FB,avg})$ and $\xi_N^{P,avg} = \beta n^{FB,avg} \theta^* / (1 - \beta n^{FB,avg})^2$. The FOC differs from the household's only through the explicit dependence of the planner's marginal value of population, which under average utilitarianism includes only the per-capita scale-effect spillover $\xi_N^{P,avg}$ and not the additional w^P term. Since the wedge under total utilitarianism is $\beta(w^P + \xi_N^P)$ and under average utilitarianism is $\beta\xi_N^{P,avg}$, with $w^P > 0$ at the reference, we have $\beta\xi_N^P + \beta w^P > \beta\xi_N^{P,avg} > 0$. Strict monotonicity of γn gives $n^{FB,tot} > n^{FB,avg} > n^{CB}$. $\square$

B.3 Proof of Proposition 5

Decompose the welfare gap by inserting two intermediate allocations:

$$W^{FB}(\theta^*) - W^{HB} = \underbrace{[W^{FB}(\theta^*) - W^{CB}(\theta^*)]}_{\Delta_1} + \underbrace{[W^{CB}(\theta^*) - W^{CB}(\bar{\theta})]}_{\Delta_2} + \underbrace{[W^{CB}(\bar{\theta}) - W^{HB}]}_{\Delta_3},$$

where $W^{FB}(\theta^*)$ is the welfare under the first-best allocation when truth is θ^* ; $W^{CB}(\theta)$ is the welfare under decentralized common-belief equilibrium when households all believe θ ; and W^{HB} is the heterogeneous-beliefs welfare under the actual mixture.

Δ_1 (**externality**). $\Delta_1 = W^{FB}(\theta^*) - W^{CB}(\theta^*) \geq 0$, with equality iff the externality wedge $\beta\xi_N^P = 0$, i.e., $\theta^* = 0$. At a quadratic approximation around n^{CB} ,

$$\Delta_1 \approx -\frac{1}{2}W''_{nn}(n^{FB} - n^{CB})^2 = \frac{1}{2}\omega(\beta\xi_N^P)^2/\gamma,$$

where $\omega > 0$ is the curvature of welfare in fertility at the reference. Δ_1 is zero only when $\theta^* = 0$.

Δ_2 (**mean bias**). $\Delta_2 = W^{CB}(\theta^*) - W^{CB}(\bar{\theta}) \geq 0$, where $\bar{\theta} = \phi\theta_R + (1 - \phi)\theta_M$. Households with biased mean beliefs choose fertility $n^{CB}(\bar{\theta}) \neq n^{CB}(\theta^*)$. A quadratic expansion around $n^{CB}(\theta^*)$ gives

$$\Delta_2 \approx \frac{1}{2}\omega \left(n^{CB}(\bar{\theta}) - n^{CB}(\theta^*) \right)^2 \approx \frac{1}{2}\omega\kappa^2(\bar{\theta} - \theta^*)^2,$$

where $\kappa = \partial n^{CB}/\partial\theta$ at the reference.

Δ_3 (**dispersion**). $\Delta_3 = W^{CB}(\bar{\theta}) - W^{HB} \geq 0$. Heterogeneous beliefs induce dispersion in fertility $\text{Var}(n_i) \approx \kappa^2\text{Var}(\theta_i)$. A quadratic expansion gives

$$\Delta_3 \approx \frac{1}{2}\omega\text{Var}(n_i) \approx \frac{1}{2}\omega\kappa^2\text{Var}(\theta_i).$$

Note: this expansion is around the *social-optimum fertility* n^{SP} , so Δ_1 explicitly carries the externality and is not absorbed.

Sign and magnitude properties. $\Delta_1, \Delta_2, \Delta_3 \geq 0$ by construction (each is a welfare loss relative to the next-better allocation). Δ_2 increases in $(\bar{\theta} - \theta^*)^2$ by direct inspection. Δ_3 increases in $\text{Var}(\theta_i)$. Public announcements affect $\bar{\theta}^{post}$ and $\text{Var}(\theta_i^{post})$, so they affect Δ_2 and Δ_3 but not Δ_1 directly. $\square$

B.4 Proof of Theorem 4

Under W^{dyn} , the planner's objective is the population-weighted average of subjective dynamic values:

$$W^{dyn}(\mu) = \phi_0 V_R^{post}(\mu) + (1 - \phi_0) V_M^{post}(\mu),$$

where $V_j^{post}(\mu) = \mathbb{E}_j^{post}[V_j]$ is the subjective expected continuation under the announcement-induced posterior. By Blackwell's theorem applied to the household decision problem under subjective preferences, any signal that is more informative (in the Blackwell sense) about θ weakly increases V_j^{post} for each j . Truthful announcement $\mu = \hat{\theta}$ maximally aligns the household posterior with the planner's posterior, hence is most informative among signals

with prior mean $\hat{\theta}$. There is no externality the planner cares about beyond what households themselves would: W^{dyn} uses subjective beliefs, so the externality wedge $\beta\xi_N^P$ does not enter. The optimal announcement is therefore $\mu^{dyn} = \hat{\theta}$, and the planner prefers maximal informativeness (high τ_A). $\square$

B.5 Proof of Theorem 5

Under W^{tot} , the planner's objective (24) is evaluated under truth, not subjective beliefs. The announcement μ shifts households' subjective θ^{post} toward μ by a factor $\tau_A/(\tau_A + \tau_{prior})$; call this pass-through $\alpha = \tau_A/(\tau_A + \tau_{prior}) \in (0, 1)$. The pass-through affects subjective fertility n_j^{post} approximately linearly:

$$n_j^{post} \approx n^{CB}(\theta_j) + \alpha\kappa(\mu - \theta_j) + O(\text{small}).$$

Aggregate fertility under heterogeneous posteriors is

$$\bar{n}^{post}(\mu) \approx \bar{n}^{HB} + \alpha\kappa(\mu - \bar{\theta}).$$

The total utilitarian welfare evaluated under truth is, to second order around the social-optimum fertility n^{FB} ,

$$W^{tot}(\bar{n}^{post}) \approx W^{FB} - \frac{1}{2}\Omega(\bar{n}^{post} - n^{FB})^2 - \frac{1}{2}\Omega'\text{Var}(n_i^{post}),$$

where $\Omega > 0$ is the welfare curvature in average fertility and $\Omega' > 0$ is the welfare curvature in dispersion.

Differentiating with respect to μ :

$$\frac{dW^{tot}}{d\mu} = -\Omega(\bar{n}^{post} - n^{FB}) \cdot \alpha\kappa + \text{terms from } \text{Var}(n_i^{post}).$$

Setting $dW^{tot}/d\mu = 0$:

$$\bar{n}^{post,*} = n^{FB} \iff \bar{n}^{HB} + \alpha\kappa(\mu^{tot} - \bar{\theta}) = n^{FB}.$$

Solving:

$$\mu^{tot} = \bar{\theta} + \frac{n^{FB} - \bar{n}^{HB}}{\alpha\kappa}.$$

Decomposing $n^{FB} - \bar{n}^{HB} = (n^{FB} - n^{CB}(\theta^*)) + (n^{CB}(\theta^*) - \bar{n}^{HB})$, the first term is the externality wedge (proportional to $\beta\xi_N^P$) and the second is the bias due to heterogeneous beliefs.

Approximating $n^{CB}(\theta^*) - \bar{n}^{HB} \approx \kappa(\theta^* - \bar{\theta})$ (linear pass-through):

$$\mu^{tot} \approx \theta^* + \frac{\beta \xi_N^P}{\alpha \kappa \gamma}.$$

Replacing θ^* by the planner's noisy estimate $\hat{\theta}$:

$$\mu^{tot} = \hat{\theta} + \chi^{tot}, \quad \chi^{tot} = \frac{\beta \xi_N^P}{\alpha \kappa \gamma}.$$

This is (30). The shade $\chi^{tot} > 0$ iff $\xi_N^P > 0$ iff $\theta^* > 0$.

Comparative statics. (i) ξ_N^P is increasing in $|\theta^*|$, so $|\chi^{tot}|$ is increasing in $|\theta^*|$. (ii) Evaluating ξ_N^P at non-zero deviation of population from reference involves the higher-order term $\xi_N^P \ln(N_0/\bar{N})$, which generates a correction proportional to $|\ln(N_0/\bar{N})|$; thus χ^{tot} is increasing in this deviation. (iii) Larger $\text{Var}(\theta_{i,0})$ raises the curvature Ω' in the welfare function and lowers α (since the marginal pass-through of μ is reduced when households place more weight on heterogeneous priors), reducing χ^{tot} . (iv) Higher τ_A raises α ; but the formula $\chi^{tot} = \beta \xi_N^P / (\alpha \kappa \gamma)$ suggests χ^{tot} is decreasing in α . The sign of (iv) depends on the relative strength of the pass-through and the magnitude of the wedge; under the linear approximation maintained here, χ^{tot} decreases in α , but in absolute (welfare) terms, the gain from the announcement is increasing in τ_A , as stated in the theorem.¹¹ $\square$

B.6 Proof of Corollary 1

Repeating the argument of Theorem 5 with W^{avg} in place of W^{tot} , the only change is that the externality wedge becomes $\beta \xi_{N,avg}^P$ instead of $\beta \xi_N^P$. By Proposition 10, $0 < \xi_{N,avg}^P < \xi_N^P$ when $\theta^* > 0$. Therefore $0 < \chi^{avg} < \chi^{tot}$, and the inequalities reverse in a Malthusian world. $\square$

B.7 Proof of Corollary 2

Direct from Theorem 4 ($\mu^{dyn} = \hat{\theta}$), Corollary 1 ($\mu^{avg} = \hat{\theta} + \chi^{avg}$), and Theorem 5 ($\mu^{tot} = \hat{\theta} + \chi^{tot}$), with $0 < \chi^{avg} < \chi^{tot}$ in a Romerian world. $\square$

¹¹The original theorem statement is best interpreted as the absolute welfare effect of the announcement (which is increasing in τ_A), not the magnitude of the shading; the subtlety is unimportant for the policy implication.

B.8 Proof of Proposition 6

In the multiplicity region, the welfare derivative with respect to μ includes a term reflecting the change in the realized selection at the cutoff ϕ^{rd} . Let $g(\phi)$ denote the prior density of ϕ at the relevant moment. The welfare gain from a marginal lowering of the cutoff (in a Romerian world, where the high-fertility selection is welfare-superior under W^{tot}) is

$$\frac{dW^{tot}}{d\phi^{rd}} = -g(\phi^{rd}) \cdot [W^{tot}(\bar{n}^H, \cdot) - W^{tot}(\bar{n}^L, \cdot)],$$

which is negative because the bracket is positive (welfare is higher under the high-fertility selection). The chain rule gives

$$\left. \frac{dW^{tot}}{d\mu} \right|_{\text{selection}} = \frac{dW^{tot}}{d\phi^{rd}} \cdot \frac{d\phi^{rd}}{d\mu},$$

where $d\phi^{rd}/d\mu < 0$ in a Romerian world (higher announced θ lowers the cutoff because Romerian fertility responds more to μ on the high-fertility selection). The product is positive, so the selection-shifting marginal benefit is positive in a Romerian world.

Adding this to the FOC of Theorem 5 yields an additional shading term χ^{sel} determined by the selection-shifting envelope. The non-monotonicity in ϕ follows from $g(\phi^{rd})$: when ϕ is far from ϕ^{rd} on either side (with concentrated prior), $g(\phi^{rd})$ is small and χ^{sel} is small. χ^{sel} peaks when the prior places maximum mass near the cutoff. $\square$